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INSTITUT ZA NAUČNO ISTRAŽIVAČKI RAD I RAZVOJ

Bosnia and Herzegovina was the fourth country in Europe that developed National version of HeartScore program !

Bosna i Hercegovina je bila četvrta zemlja u Evropi koja je razvila Nacionalnu verziju HeartScore programa !



Bosnia and Herzegovina version of HeartScore is developed on the languages of the people of Bosnia and Herzegovina i.e. Bosnian, Serbian and Croatian! Program is easy to use and accessible at www.heartscore.org/eu !

Verzija za Bosnu i Hercegovinu razvijena je na jezicima naroda Bosne i Hercegovine, Bosanskom, Srpskom i Hrvatskom ! Program je jednostavan za upotrebu preko web stranice www.heartscore.org/eu !

Bosnia Herzegovina	France	Russian Federation
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New Central Medical Building - University Clinical Center Sarajevo
Novi Centralni Medicinski Blok - Univerzitetski klinički centar u Sarajevu





Novi Evropski vodič za prevenciju tromboembolizma kod A Fib

CHA_2DS_2-VASc skor za procjenu rizika od tromboembolizma kod A Fib!

Risk factor-based point-based scoring system - CHA_2DS_2 -VASc

Risk factor	Score
Congestive heart failure/LV dysfunction	1
Hypertension	1
Age ≥75	2
Diabetes mellitus	1
Stroke/TIA/thrombo-embolism	2
Vascular disease*	1
Age 65–74	1
Sex category (i.e. female sex)	1
Maximum score	**9**

*Prior myocardial infarction, peripheral artery disease, aortic plaque. Actual rates of stroke in contemporary cohorts may vary from these estimates.

Major i non-major riziko fakori za procjenu tromboembolizma kod A Fib!

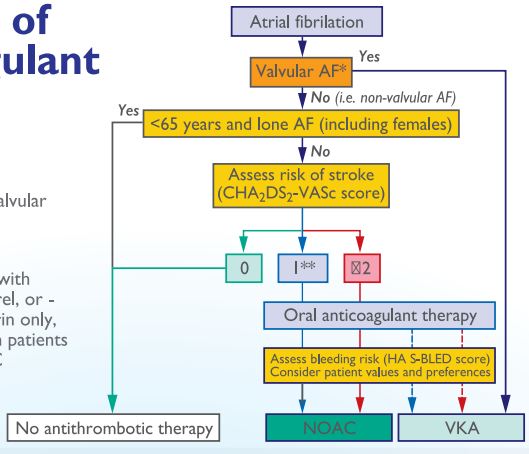
Risk factors for stroke and thrombo-embolism in non-valvular AF

Major risk factors	Clinically relevant non-major risk factors
Previous stroke	CHF or moderate to severe LV systolic dysfunction [e.g. LV EF ☒ 40%]
TIA or systemic embolism	Hypertension
Age ☒75 years	Diabetes mellitus
	Age 65-74 years
	Female sex
	Vascular disease

AF = atrial fibrilation; EF = ejection fraction (as documented by echocardiography, radio nuclide ventriculography, cardiac catheterization, cardiac magnetic resonance imaging, etc.); LV = left venticular; TIA = trasient ischaemic attack.

Algoritam antikoagulantne terapije nakon procjene CHA_2DS_2VASc i major risk faktora!

Choice of Anti-coagulant



* Includes rheumatic valvular AF, hypertrophic cardiomyopathy, etc.

** Antiplatelet therapy with aspirin plus clopidogrel, or - less effectively - aspirin only, may be considered in patients who refuse any OAC

NOAC - Novel Oral Anticoagulants, VKA - Vitamin K Antagonists

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AIMS AND SCOPE

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Association of anti-cyclic citrullinated peptide antibodies and C-reactive protein in patients with rheumatoid arthritis

Povezanost antitijela na ciklične citrulinirane peptide i C-reaktivnog proteina kod pacijenata sa reumatoidnim artritisom

Nudžejma Veladžić*, Hana Smajlović, Ena Šaranović, Anita Risonjić, Fatima Veladžić, Amela Dervišević, Nermina Babić, Nesina Avdagić

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ABSTRACT

Rheumatoid arthritis (RA) is an independent risk factor for adverse cardiovascular events accounting for a significant proportion of mortality among these patients. C-reactive protein (CRP) has been indicated as independent risk factor in a variety of cardiovascular disease (CVD). Anti-cyclic citrullinated peptide (anti-CCP) antibodies are associated with higher frequency of extra-articular manifestations and poorer outcomes in RA patients. The aim was to evaluate association between anti-CCP antibodies and CRP and to examine whether such an association is present when duration of RA is taken into account. A total of 84 RA patients of both genders were divided in two groups based on the disease duration: Group RA1 (n=55) < 48 months and Group RA2 (n=29) ≥ 48 months. Concentration of anti-CCP antibodies was higher, but not statistically significant in Group RA2 compared to Group RA1 (p=0.41). Patients in Group RA2 had significantly higher CRP concentration compared to patients in Group RA1 (p=0.022). Significant positive correlation was found between concentration of anti-CCP antibodies and CRP in Group RA2 (Rho=0.511; p=0.005), while positive, although not significant, correlation was observed between these parameters in Group RA1 (Rho=0.178; p=0.201). RA patients with prolonged disease duration have high concentration of anti-CCP antibodies and CRP and therefore should be evaluated for CVD more meticulously.

Key words: rheumatoid arthritis, anti-CCP antibodies, C-reactive protein, cardiovascular disease

SAŽETAK

Reumatoidni artritis (RA) je nezavisni faktor rizika za razvoj kardiovaskularnih bolesti (KVB). C-reaktivni protein (CRP) je procijenjen kao nezavisni faktor u procjeni rizika od nastanka KVB. Antitijela na ciklične citrulinirane peptide (anti-CCP antitijela) su povezana sa većom učestalošću vanzglobnih manifestacija i lošijim ishodom kod pacijenata sa RA. Cilj je bio ispitati povezanost anti-CCP antitijela i CRP i utvrditi da li na navedenu povezanost utječe dužina trajanja bolesti. 84 RA pacijenata, oba spola, su na osnovu dužine trajanja bolesti bili podjeljeni na dvije grupe: Grupa RA1 (n=55) < 48 mjeseci i Grupa RA2 (n=29) ≥ 48 mjeseci. Koncentracija anti-CCP antitijela je bila viša, ali ne značajno u Grupi RA2 u poređenju sa Grupom RA1 (p=0.410). Pacijenti u Grupi RA2 su imali značajno veću serumsku koncentraciju CRP u poređenju sa pacijentima u Grupi RA1 (p=0.022). Ustanovljena je značajna pozitivna korelacija između koncentracije anti-CCP antitijela i CRP u Grupi RA2 (Rho=0.511; p=0.005) te pozitivna, mada ne značajna, korelacija između ovih parametara u Grupi RA1 (Rho=0.178; p=0.201). RA pacijenti kod koji bolest duže traje imaju visoku serumsku koncentraciju anti-CCP antitijela i CRP i stoga bi trebali biti podvrgnuti detaljnijem ispitivanju s ciljem pravovremene dijagnoze mogućih KVB.

Ključne riječi: reumatoidni artritis, anti-CCP antitijela, C-reaktivni protein, kardiovaskularne bolesti

INTRODUCTION

Rheumatoid arthritis (RA) is the most common autoimmune inflammatory polyarthritis (1). About 1% of world population has this chronic systemic inflammatory disease of unknown cause. An external trigger (e.g. cigarette smoking, infection or trauma) that triggers an autoimmune reaction, leading to synovial hypertrophy and chronic joint inflammation along with the potential for extra-articular manifestations, is theorized to occur in genetically susceptible individuals. Patients with RA have an increased risk of cardiovascular disease (CVD) that may not always be associated with traditional CV risk factors such as smoking, hypertension, diabetes mellitus, being overweight and dyslipidemia (2). In RA patients, severe extra-articular manifestations are associated with higher risk of development of CV complications (3).

Anti-cyclic citrullinated peptide (anti-CCP) antibodies are potentially important surrogate marker for diagnosis and prognosis in RA (4). In RA patients, anti-CCP antibodies are associated with greater inflammatory activity, poorer radiologic outcome and higher frequency of extra-articular manifestations. The anti-CCP response may also be involved in the pathogenesis of RA, both because of the presence of antibodies directed against citrullinated proteins in the target tissue and the association between anti-CCP antibodies and severe RA (5).

Anticyclic citrullinated peptide (anti-CCP) antibodies are associated with the pathogenesis, clinical expression, and cardiovascular risk in RA (6). By immunehistochemistry Sokolove et al. (7) demonstrated the presence of citrullinated proteins within the damaged endothelium in atherosclerotic plaque. RA patients positive for anti-CCP or rheumatoid factor (RF) have more endothelial dysfunction (7). So far, it has been shown that positivity of anti-CCP antibodies is associated with the increased carotid intima-media thickness (cIMT) which is marker of atherosclerosis (8-12).

C-reactive protein (CRP) is a liver-derived pattern recognition molecule that is increased in inflammatory states. It rapidly increases within hours after tissue injury and it is suggested that it is part of the innate immune system and contributes to host defence. Since CVD is at least in part an inflammatory process, CRP has been investigated as a predictor of cardiovascular mortality in the general population. CRP is a strong independent risk factor for CVD that adds to the predictive value of risk models based on usual factors alone (13).

The aim of our study was to evaluate association between anti-CCP antibodies and CRP in patients with RA and to examine whether such an association is present when duration of RA is taken into account.

MATERIALS AND METHODS

The cross-sectional study included 84 RA patients of both genders (78 females and 6 males). Diagnosis of RA was determined by a rheumatologist based on anamnesis, clinical findings, laboratory tests and radiological examinations. Based on disease duration RA patients were divided into two groups: Group RA1 (n=55): patients with RA duration of less than 48 months and Group RA2 (n=29): patients with RA duration of 48 or more than 48 months. All subjects involved in the study went through detailed anamnestic questionnaire, medical history, physical examination and standard laboratory analyses. Exclusion criteria included other rheumatic diseases or any local or systemic illness caused by bacterial and viral infections and pregnancy. Written informed consent was obtained from all subjects. The study was carried out at the Cardiology Clinic of the University Clinical Center Sarajevo (UCCS), in the period between 24 February to 27 June 2015, with an approval from the Ethics Committee of the UCCS. Investigations were carried out in accordance with the Declaration of Helsinki as revised in 2000.

Blood sampling

Blood samples for anti-CCP antibodies and CRP concentrations were taken from the cubital vein using vacutainer technique.

Blood chemistry analysis

Serum anti-CCP antibodies levels were determined by Grifols Triturus automated enzyme-linked immunosorbent assay (ELISA) device using RA/CP-Detect Enzyme Immunoassay kit. High sensitivity CRP was determined by particle-enhanced immunonephlometry (BN System, Dade Behring, Marburg, Germany) at the Institute of Clinical Chemistry and Biochemistry of the UCCS. The lower limit of detection of this assay was 0.18 mg/L.

Statistical analysis

The Shapiro-Wilk test of normality was used to test the distribution of variables. Since all variables were skewed they were presented as median and interquartile ranges. The differences in variables between the groups were determined by the non-parametric Mann-Whitney U-test. Chi-square analysis was used for categorical variables. Correlation coefficient between variables was assessed by Spearman's test. A p value < 0.05 was considered statistically significant for all comparisons. The SPSS for Windows (version 13.0; SPSS, Chicago, IL, USA software was used.

RESULTS

The baseline characteristics of RA patients in Group RA1 and Group RA2 are presented in Table 1. There was no statistically significant difference in age and gender between RA patients in Group RA1 and Group RA2. A significant difference ($p<0.0005$) in disease duration was found between the study groups.

Table 1 Baseline characteristics of the patients in the Group RA1and Group RA2.

	Group RA1 (n=55)	Group RA2 (n=29)	p
Ages (years)	60.0 (52.0-63.0)	56.0 (48.0-61.5)	NS
Disease duration (month)	13.0 (8.9-20.0)	60.0 (52.0- 80.0)	< 0.0005
Male/female (%)	5/50 (9.1/90.9)	1/28 (3.4/96.6)	NS

Values are presented as median (25th and 75th percentiles); n (%); NS - not significant; p - probability

The median and interquartile ranges of anti-CCP antibodies serum level in Group RA2 5.6 U/ml; (2.4-6.8 U/ml) was higher in comparison to Group RA1 5.0 U/ml (2.5-6.2 U/ml), but the difference was not statisticly significant (p=0.41) (Figure 1).

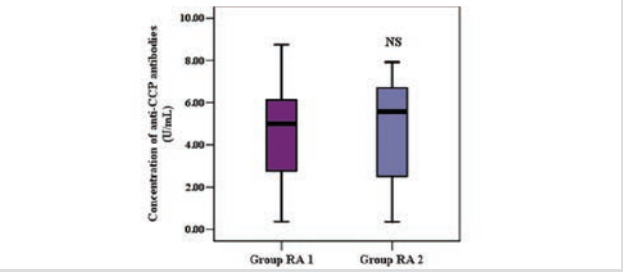


Figure 1 Serum anti-CCP antibodies level in RA patients according to disease duration.

Patients in Group RA2 had significantly higher (p=0.022) median serum CRP concentration 20.4 mg/L (10.4-46.3 mg/L) compared to median serum CRP concentration in Group RA1 3.2 mg/L (7.8-29.5 mg/L) (Figure 2).

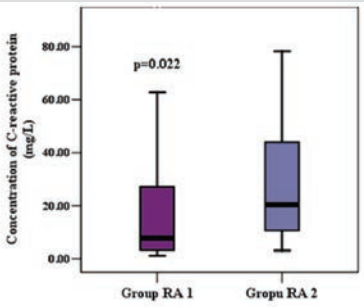


Figure 2 Serum CRP level in RA patients according to disease duration.

Positive, although not significant (Rho=0.178; p=0.201), correlation was observed between concentration of anti-CCP antibodies and CRP in Group RA1 (Figure 3).

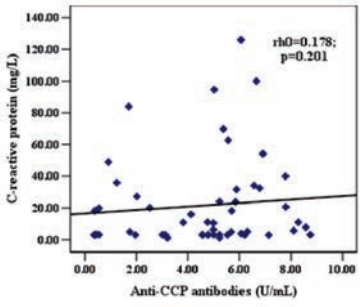


Figure 3 Correlation between anti-CCP antibodies and CRP in Group RA1.

There was statistically significant positive correlation between the concentration of anti-CCP antibodies and CRP (Rho=0.511; p=0.005) in Group RA2 (Figure 4).

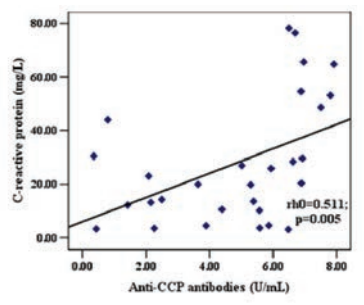


Figure 4 Correlation between anti-CCP antibodies and CRP in Group RA2.

DISCUSSION

Mortality risk in RA patients is 1.5 higher than in general population and occurs largely as a result of higher rates of cardiovascular death (14). Evidences support the hypothesis that chronic inflammation could increase cardiovascular risk in patients with RA (15). Patients with RA have elevated levels of the acute-phase reactant, CRP, a marker of inflammation associated with increased cardiovascular risk (16). Moreover, RA patients with elevated erythrocyte sedimentation rate (ESR) have a higher rate of cardiovascular death than those without elevated ESR (17). Different authors suggested that the higher prevalence of cardiovascular events in RA patients could be explained by subclinical atherosclerosis and its relationship with inflammatory process. Atherosclerosis is now considered as an inflammatory autoimmune disease in which immune system plays an important role (18). Patients with RA who are rheumatoid negative or anti-cyclic citrullinated peptide negative have less atherosclerotic disease than patients with RA who have these autoantibodies (18).

Our studies are consistent with numerous studies based on correlation between anti-CCP antibodies and CRP with cardiovascular risk.

The results of our study showed that anti-CCP antibodies serum levels were higher in group RA 2 compared to the group RA 1, but the difference was not statisticlly significant (p=0.41). This result actually confirms the theses about association between anti-CCP antibodies and inflamatory activity, poorer radiologic outcome as well as higher frequency of extra articular manifestations (5) which were all characteristic for patients in group RA 2.

Patients with disease duration longer than 48 months had significantly higher serum CRP concentration than patients whose disease duration was shorter than 48 months which was a sign of immune system host defence. Since cardiovascular diseases are part of an inflammatory process this result show that there was higher incidence of cardiovascular diseases in patients with disease duration longer than 48 months.

There were positive correlations between anti-CCP antibodies and CRP in both groups which leads to a conclusion that analysing anti-CCP antibodies values in serum can be helpful in evaluation of cardiovascular disease.

Lopes-Longo et al. (19) investigated association between anti-cyclic citrullinated peptide antibodies and ischemic heart disease in patients with RA. They included 937 patients with RA and found that there was no association between the anti-CCP antibodies and cardiovascular risk factors such as smoking, hypertension, dyslipidemia, being overweight or diabetes mellitus. However, patients who had positive anti-CCP antibodies experienced more frequent ischemic heart disease and had higher mortality rates. Their results match ours showing that anti-CCP antibodies in patients with RA is independently associated with the development of ischemic heart disease.

Arnab et al. (20) included eighty patients with RA having disease duration of at least 3 years in their hospital-based, cross-sectional and observational study. They wanted to determine the role of anti-CCP antibody as an independent risk factor for developing CV complications as documented by carotid intima medial thickness and abnormal echocardiography in established RA patients. This study stressed the important role of anti-CCP antibody in myocardial dysfunction due to inflammation in RA patients. Both atherosclerotic vascular involvement and cardiac abnormalities including pericardial, myocardial and endocardial involvements were higher among anti-CCP positive RA patients. They concluded that patients with high titter of anti-CCP antibody were associated with prolonged disease duration and increased disease activity should be evaluated for CV morbidity in more detail.

Vazquez Del-Mercado, et al. (21) conducted a study which goal was to determine whether serum levels of anti-CCP antibodies and

CRP were associated with increased carotid intima media thickness. This was a cross-sectional analysis of a cohort of rheumatoid arthritis patients without cardiovascular risk factors. Study included 45 anti-CCP positive and 37 anti-CCP negative RA patients and 62 healthy controls. The carotid intima media thickness significantly ($p<0.001$) positive correlated with levels of anti-CCP antibodies, CRP, TNFα and IL-6 serum levels. In multiple regression analysis, cIMT was associated with CRP ($p<0.001$) and anti-CCP levels ($p=0.03$). They concluded that levels of anti-CCP antibodies and CRP were associated with increased carotid intima media thickness and cardiovascular risk supporting a clinical role of the measurement of carotid intima media thickness in RA in predicting and preventing cardiovascular events. Positive correlations between anti-CCP antibodies and CRP with carotid intima media thickness speak about influence of these markers on cardiovascular diseases and that is why we find results of this study to be consistent with ours.

Mackey et al. (22) did a huge study on over 160.000 postmenopausal women. They wanted to evaluate the incidence of cardiovascular disease (CVD) morbidity and mortality over the course of 10 years among more than 160.000 postmenopausal women in the Women's Health Initiative (WHI) in relation to self-reported rheumatoid arthritis (RA), taking disease-modifying antirheumatic drugs (DMARDs), anti-cyclic citrullinated peptide (anti-CCP) positivity, rheumatoid factor (RF) positivity, CVD risk factors, joint pain and inflammation (white blood cell count and interleukin-6 levels). Among women with RA, anti-CCP positivity and RF positivity were not significantly associated with higher risk of any outcomes, despite slightly higher risk of death for those who were anti-CCP positive than for those who were anti-CCP negative. They concluded that CVD risk was strongly associated with CVD risk factors, joint pain severity and inflammation, but not with anti-CCP positivity or RF positivity. These results do not match ours.

CONCLUSION

Although our study included relatively small number of patients, our results showed positive correlation between serum concentration of anti-CCP antibodies and CRP. RA patients with prolonged disease duration had high titter of anti-CCP antibodies and CRP concentration and therefore should be evaluated for cardiovascular morbidity more meticulously.

Conflict of interest: none declared.

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Frequency of physical comorbidity in schizophrenia

Učestalost tjelesnog komorbiditeta kod shizofrenije

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ABSTRACT

Schizophrenia is a chronic mental illness with a relatively high frequency of about 1% in the general population. It is most common in the early adolescence, but can also occur in middle and older age. Health professionals who work with patients should be more informed about the increased risk of somatic diseases in these patients. The aim of this study is to determine the presence of somatic diseases in schizophrenic patients and the relation of physical comorbidity with frequent rehospitalization.

Key words: schizophrenia, physical comorbidity, frequency, hospitalization

SAŽETAK

Shizofrenija je hronična duševna bolest sa relativno visokom učestalošću od oko 1% u opštoj populaciji. Najčešće se javlja u ranoj adolescenciji, ali može se javiti i u srednjoj i starijoj životnoj dobi. Zdravstveni profesionalci koji rade sa pacijentima trebaju više biti obaviješteni o povećanom riziku od somatskih oboljenja kod ovih pacijenata. Cilj rada je utvrditi prisutnost somatskih oboljenja kod shizofrenih pacijenata i povezanost tjelesnog komorbiditeta sa učestalim rehospitalizacijama.

Ključne riječi: shizofrenija, tjelesni komorbiditet, učestalost, hospitalizacija

INTRODUCTION

Schizophrenia is a chronic mental illness with a relatively high frequency of about 1% in the general population. It usually occurs in early adolescence, but can also occur in middle and old age.

People with schizophrenia have a higher risk of other diseases and premature death (1), and many factors affect on it to be increased (2). Some of the important factors are: increasing of mental and physical stress, negative symptoms, loss of reality and initiatives. Regarding external factors, we must consider the following: unhealthy lifestyles, smoking, eating junk food, lack of physical activity and side effects of pharmacotherapy (3). Antipsychotics also increase the risk of developing metabolic syndrome and diabetes mellitus (4).

Health professionals who work with patients need to be more informed about the increased risk of somatic diseases in these patients (5). In addition, there is a lot of space for improvement in cooperation between primary and secondary health care (5).

The aim was to determine the:

- presence of somatic diseases in schizophrenic patients;
- relationship between somatic comorbidity and frequent rehospitalization.

MATERIALS AND METHODS

The sample was made on 100 patients diagnosed with schizophrenia and treated in the Public Health Institution "Sveti apostol Luka" in Doboj. Randomized patients of both sexes were included, over 18 years of age.

The criterion for inclusion was the diagnosis of schizophrenia set by a psychiatrist. Data on somatic comorbidity was collected on the basis of medical documentation and the confirmed diagnosis by a specialist for somatic disorders.

Questionnaire was designed to provide insight into the socio-demographic data, disease duration, presence of somatic diseases, presence of comorbidity data and rehospitalization.

The assumption was that the presence of somatic disease, as a risk factor, would affect a larger number of repeated hospitalizations.

Statistical analysis gathered the descriptive statistics and inferential statistical methods. In the area of descriptive statistics we used the average value, standard deviation and frequency. In the area of inferential methods we used the chi-square test and Pearson's chi-square test of similarities. Pearson's chi square test was used in order to find the connection between the variables, and the chi-square test for questioning the independence between two variables or coincidences.

RESULTS

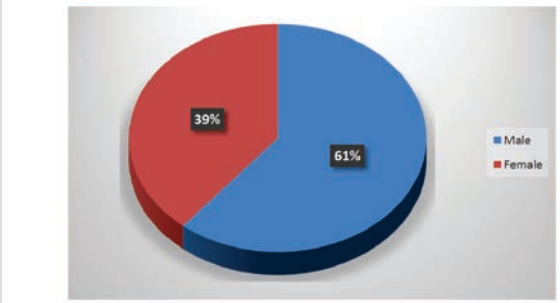


Figure 1 Distribution of patients by gender.

The results of χ2 test (χ2=4.840; p=0.028) showed that the number of statistically significant male patients (p<0.05) was higher than the number of female patients.

Table 1 Distribution of patients by age group.

Age in decades	N	%
18-29	2	2.0
30-39	11	11.0
40-49	26	26.0
50-59	37	37.0
60-100	24	24.0
Total	100	100.0

The results of χ2 test (χ2=37.300; p=0.000) showed a statistically significant difference (p<0.05) in the number of patients between the age groups. The age group of 50 to 59 years was the most dominant in the number of patients.

Table 2 Distribution of patients by professional qualifications.

Education	No school	4	4.0
	Crafts	11	11.0
	Elementary school	27	27.0
	High school	51	51.0
	University	7	7.0
	Total	100	100.0

The results of χ2 test (χ2=75.800; p=0.000) showed a statistically significant difference (p<0.05) in the number of patients in respect of their professional qualifications. The highest number of patients had secondary education, 51%.

Table 3 Distribution of patients according to marital status.

		N	%
Marital status	Divorced	21	21.0
	Married	31	31.0
	Widow / Widower	6	6.0
	Never married	42	42.0
	Total	100	100.0

Based on the results of the χ2 test (χ2=28.080; p=0.000), there was a statistically significant difference (p<0.05) according to marital status, and we noticed that the most common group of patients were those who never married, with a share of 42%.

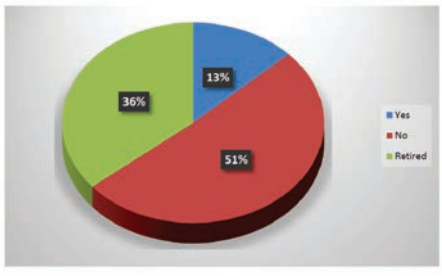


Figure 2 Distribution of patients with respect to employment.

The results of χ2 test (χ2=21.980; p=0.000) showed a statistically significant difference (p<0.05) in the number of patients in relation to employment. Unemployed patients were the most frequent and accounted for 51% of the patients included in the study.

Table 4 Distribution of patients based on the length of treatment.

		N	%
The duration of illness	Up to 5 years	16	16.0
	From 6 to 10 years	32	32.0
	From 11 to 20 years	37	37.0
	More than 20 years	15	15.0
	Total	100	100.0

The results of χ2 test (χ2=14.960; p=0.002) showed a statistically significant difference (p<0.05) in the number of patients based on the duration of the disease. The largest number of patients was treated for 11 to 20 years and the share was 37%.

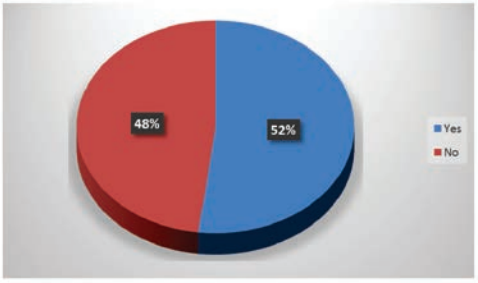


Figure 3 Presence of somatic illness.

The results of χ2 test (χ2=0.160; p=0.689) showed no statistically significant differences (p>0.05) between the groups of patients with and without somatic diseases, and therefore we come to a conclusion that somatic diseases are often present in patients suffering from schizophrenia.

Table 5 Distribution of patients with somatic disease by sex.

			SEX		Total
			Male	Female	
Somatic diseases	Yes	N	26	26	52
		%	42.6	66.7	52.0
	No	N	35	13	48
		%	57.4	33.3	48.0
Total		N	61	39	100
		%	100.0	100.0	100.0

Based on the results of the Pearson χ2 test (χ2=5.510; p=0.019), it was concluded that there was a statistically significantly higher (p<0.05) presence of somatic illnesses in female than in male patients.

Table 6 Rehospitalization of patients with somatic diseases.

			Rehospitalization		Total
			Yes	No	
Somatic diseases	Yes	N	31	21	52
		%	67.4	38.9	52.0
	No	N	15	33	48
		%	32.6	61.1	48.0
Total		N	46	54	100
		%	100.0	100.0	100.0

Based on the results of the Pearson χ2 test (χ2=8.085; p=0.004), there was a statistically significant (p<0.05) correlation between the

presence of somatic diseases and rehospitalization. The presence of somatic illness was positively correlated with number of rehospitalizations.

Table 7 Correlation of comorbidity and rehospitalization.

			Rehospitalization		Total
			Yes	No	
Comorbidity	Yes	N	37	23	60
		%	80.4	42.6	60.0
	No	N	9	31	40
		%	19.6	57.4	40.0
Total		N	46	54	100
		%	100.0	100.0	100.0

The results of the Pearson $\chi 2$ test ($\chi 2=14.822$; $p= 0.000$) showed a statistically significant ($p<0.05$) association between comorbidity and rehospitalization. Rehospitalization was more prevalent in patients with present comorbidity.

Table 8 The most common somatic diseases in male and female patients.

Somatic disease		M	F	TOTAL
Hypertensio arterialis	N	11	16	27
	%	40.7	59.3	100.0
Hyperlipidemia	N	10	6	16
	%	62.5	37.5	100.0
Diabetes mellitus	N	5	5	10
	%	50.0	50.0	100.0
Other diseases	N	12	18	30
	%	40.0	60.0	100.0

Based on the results of the Fisher's exact test ($p=0.471$) we came to a conclusion that there was no statistically significant difference ($p>0.05$) between male and female patients in respect to somatic diseases.

DISCUSSION

Quran et al. (1) estimated that 45% of patients in the public mental health system in California had somatic diseases, while in our study the presence of somatic diseases was 52%. Statistically significant part of these diseases can cause or worsen the existing condition of patients. Similar results were found in the study of the Psychiatric Clinic Koranyi (2): 43% of patients had somatic diseases, and out of these, 46% were diagnosed by specialists of other medical fields. According to Luijks (3) 84% of patients had some sort of somatic disease. It was found that 27.2% of patients had three or more chronic somatic diseases, with cardiovascular diseases being the most common.

Balf et al. (4) found that 46% of patients who participated in the survey had undiagnosed somatic diseases that caused or contributed to the worsening of psychiatric illness; 80% had somatic diseases that require treatment; 4% had premalignant lesions.

According to Nuts et al. (7), the presence of diabetes in patients with schizophrenia was 8% and in our study it was 10%. The presence of hypertension was 22.6%, and in our study it was 27%, while in the study by Daniel J. Smith (8) the presence of hypertension was significantly higher, 95%. Hyperlipidemia was present in 85% of patients, and in our study it was present in 16%. They found that the rate of comorbidity in patients with schizophrenia was 58.4%, while the rate of comorbidity in our study was slightly lower.

CONCLUSION

The results of this study do not significantly differ from other researchers. Somatic comorbidity often develops in patients with schizophrenia. Patients with comorbid disease are hospitalized more often than those without somatic diseases. Previous analysis showed that the number of patients who had a somatic disease was statistically the same as the number of patients who did not have a somatic disease, therefore, there is an equal probability that a patient will or will not have a somatic disease. Therefore, it was confirmed that somatic diseases are often present in schizophrenic patients. Patients who had somatic comorbidities accounted for 80.4% of the total number of rehospitalized people, and patients who did not have somatic comorbidities accounted for only 19.6% of the total number of rehospitalized patients. This confirmed the hypothesis that somatic comorbidity contributes to an increased number of rehospitalization. This statistical analysis shows that the number of patients with somatic diseases is statistically significantly higher than the number of those without somatic diseases in a group of rehospitalized patients. The presence of somatic diseases in schizophrenic patients increases with the number of rehospitalization.

Conflict of interest: none declared.

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Complications of central venous catheterization at the Pediatric Intensive Care Unit

Komplikacije primjene centralnih venski kateter u Jedinici pedijatrijskog intenzivnog liječenja

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ABSTRACT

Application of intravascular catheters for the purpose of venous approach and hemodynamic monitoring of patients has become a central part of contemporary medicine, especially in intensive care units. Complications which occur during the use of central venous catheters can be divided into: acute, subacute and chronic. Infection is the most serious complication of central venous catheterization. Other complications relate to malposition of catheter, artery puncture, hematoma, hemathorax, pneumothorax, thrombosis and embolism. The aim of the study was to present our experiences with central venous catheterization. A total of 1246 central venous catheters were placed in the period from March 2009 to March 2016. Out of the total number of central venous access 326 (26.1%) was performed via the internal jugular vein, 561 (45%) via the subclavian vein, and 359 (28.8%) via the femoral vein. Out of the total number of acute compilations, 57(4.57%) related to artery puncture, 60 (4.81%) to hematoma, 5 (0.4%) to hemathorax, 8 (0.35%) to pneumothorax, 6 (0.48%) to trombosis, and 15 (1.2%) to malposition of catheter. Infection as a chronic complication was proven in 36 (2.9%) cases. The leading agents were gram-negative bacteria registered in 22 (1.76%) cases of which the most common was Klebsiella pneumoniae.

Key words: central venous catheter (CVK), complications

SAŽETAK

Primjena intravaskularnih katetera zbog krvnožilnog pristupa i hemodinamskog monitoriranja bolesnika postala je središnji dio suvremene medicine, osobito u jedinicama intenzivnog liječenja. Komplikacije koje nastaju prilikom korištenja centralnih venskih katetera dijele se na: akutne, subakutne i kronične. Najozbiljnija komplikacija koja nastaje zbog upotrebe centralnih venskih katetera je infekcija. Ostale komplikacije su: malpozicija katetera, punkcija arterije, hematom, hematotoraks, pneumotoraks, tromboze i embolije. Cilj rada je prikazati naša iskustva sa centralnim venskim kateterima. U periodu ožujak 2009. - ožujak 2016. godine plasirano je ukupno 1246 centralnih venskih katetera. Od toga u v.jugularis int 326 (26,1%), v.subclavia 561 (45%), v.femoralis 359 (28,8%). Od akutnih komplikacija: punkcija arterije ukupno 57 (4,57%), hematom 60 (4,81%), hematotoraks 5 (0,4%), pneumotoraks 8 (0,35%), tromboza 6 (0.48%), malpozicija 15 (1,2%). Infekcija kao kronična komplikacija dokazana je kod 36 (2,9%) pacijenata. Vodeći uzročnici su gram negativne bakterije 22 (1,76%) od kojih najčešće Klebsiella pneumoniae.

Ključne riječi: centralni venski kateter (CVK), komplikacije

INTRODUCTION

Intravascular catheters are imperative in contemporary everyday medical practice, especially in the intensive care units (1). Once placed, they enable necessary venous access. However, their application carries a risk of acute, subacute and chronic complications (2).

The infection occurs as local complication on the spot of the catheter placement, but also as sepsis, septic thrombophlebitis, endocarditis and other metastatic infections (lung abscess, brain abscess, osteomyelitis, endophthalmitis) (3). The majority of these complications is related to catheterization of patients at the intensive care unit, where central venous access is required through a longer period of time, where patients are colonized with hospital microorganisms, and the catheter is often frequently manipulated for the application of liquid, medications, and blood derivates.

In case of an urgent catheter placement, it is occasionally difficult to enable aseptic placement conditions. Some catheters are frequently manipulated during the day due to hemodynamic measuring or taking samples for the purpose of laboratory tests, which increase the possibility of contamination and infection (4).

Other complications include: malposition of catheter, artery puncture, hematoma, hemathorax, pneumothorax, thrombosis and embolism (5,6).

Approximately 200 CVK are placed annually at the Pediatric Intensive Care Unit of the CCUS Pediatric Clinic.

Indications for the central venous characterization are: long-term

systematic application of drugs, continuous application of cardiovascular drugs and hyperosmolar fluids, need for frequent blood taking for laboratory tests, invasive monitoring of cardiovascular functions, long-term full and partial parenteral nutrition, hemodialysis.

The catheterization is performed by a Pediatrician - practicing at the Intensive Care Unit. For that purpose access via the subclavian and femoral vein are the most frequent, whereas assess via the internal jugular vein is less frequent. The selection of thee big vein depends of several factors such as: age group, patient's condition and anatomical structure, local findings and the physician's experience.

The Seldinger technique was used for a catheter insertion. The position of the inserted catheter was controlled by ultrasound and x-ray if necessary (7,8).

Aim of the study

The aim of the study was to present personal observations and experiences with acute venous access (central venous catheters) in the treatment of patients hospitalized at the Intensive Care Unit.

MATERIALS AND METHODS

In the period from March 2009 to March 2016 a total of 1246 catheters were placed at the Pediatric Intensive Care Unit of the CCUS Pediatric Clinic. Occurrence of acute and chronic complications was monitored depending on the type of the central venous catheter (internal jugular vein, subclavian vein, femoral vein).

RESULTS

A total of 1246 central venous catheters were placed in the period from March 2009 to March 2016; catheterization via the subclavian vein was performed in 561 (45%) patients, via the femoral vein in 359 (28.8%) patients, and in 326 (26.1%) patients via the jungular vein (9).

Table 1 Complications during CVC placement.

	Internal jugular vein	Subclavian vein	Femoral vein	Total	
Artery puncture	32 (2.5%)	13 (1.04%)	12 (0.96%)	57 (4.57%)	
Hematoma	34 (2.7%)	13 (1.04%)	13 (1.04%)	60 (4.81%)	
Hemathorax	1 (0.08%)	4 (0.32%) 1 drainage	0	5 (0.4%)	
Pneumothorax (partial)	3 (0.24%)	5 (0.4%)	0	8 (0.35%)	
Thrombosis	2 (0.165)	0	4 (0.32%)	6 (0.48%)	
Malposition	2 (0.16%)	10 (0.8%)	3 (0.24%)	15 (1.2%)	
Total	74 (5.9%)	45 (3.6%)	32 (2.56%)	117 (9.4%)	34 (2.7%)

Out of the total number of complications that occurred during the placement and procedures with central venous catheters, there were 117 acute complications of which 57 (4.57%) related to artery puncture, frequently during the placement of the central venous catheter into the internal jugular vein in 32 (2.5%) patients, into the subclavian vein in 13 (1.04%) patients and the least complications occurred during the placement of the central venous catheter into the femoral vain, in 12 (0.96%) patients. Hematoma as an acute complication was present in 60 (4.81%) patients; in 34 (2.7%) patients it occurred on the occasion of the jugular vein access, and in 13 (1.09%) patients during the subclavian and femoral vein access respectively (10,11,12).

The most common subacute complication was malposition of the placed catheter which occurred in 10 (0.8%) patients on the occasion of catheterization via the subclavian vein; in 3 (0.24%) patients on the occasion of catheterization via the femoral vein and in 2 (0.16%) patients on the occasion of the jugular vein catheterization (6).

Out of other subacute complications pneumothorax was recorded in a total of 8 (0.35%) patients, in 5 (0.4%) patients during the subclavian vein access and in 3 (0.24%) patients during the internal jugular vein access. Hemothorax as a complication most frequently occurred during the subclavian vein access, in 4 (0.32%) patients, and in one patient (0.08%) during the internal jugular vein access (2).

Thrombosis as a chronic complication frequently occurred during the femoral vein access, in 4 (0.32%) patients and in 2 (0.16%) patients during the internal jugular vein access. Thrombosis as a complication of the subclavian vein catheterization was not recorded (13).

Table 2 Infections related to CVC.

Gram-negative infections (Kl.pneumoniae, Acinetobacter baumannii, Pseudomonas spp et aeruginosa, Stenotrophomonas maltophiliae)	22 (1.76%)
Gram-positive infections (Staph. Epidermidis, Staph. Aureus)	11 (0.9%)
Mycotic infections (Candida)	3 (0.24%)

Infection, as the most serious complication related to the placement and presence of the central venous catheter, was recorded in a total of 36 (2.9%) patients. The largest number of infections was caused by gram-negative bacteria (Kl.pneumoniae, Acinetobacter baumannii, Pseudomonas spp et aeruginosa, Stenotrophomonas maltophiliae) - in a total of 22 (1.76%) patients (14). Infections caused by gram-positive bacteria (Staphylococcus epidermidis and St. aureus) were recorded in 11 (0.9%) patients. Mycotic infections caused by Candida albicans were the least recorded, occurring in only 3 (0.24%) patients (15).

DISCUSSION

Placement of the central venous catheter is part of a daily medical practice at the Pediatric Intensive Care of the CCUS Pediatric Clinic. The most common indications for the central venous characterization are: lack of necessary vein access, long-term full or partial parenteral nutrition, long-term systematic application of drugs, continuous application of cardiovascular drugs and hyperosmolar fluids or medication with non-physiological pH (Dopamine, Phenobarbitan, Vancomycin), need for frequent blood taking for laboratory tests, invasive monitoring of cardiovascular functions, hemodialysis.

Absolute contraindications include: damages or venous diseases and/or congenital vascular malformations; whereas relative contraindications include: acute inflammatory skin disease above the planned puncture spot, bacteremia, coagulation disturbance and thrombocytopenia.

Subclavian vein is the most commonly used blood-vein access

(45%). The Seldinger technique is used for a catheter insertion. Position of the placed catheter is controlled by ultrasound and x-ray if needed.

The most common acute complication was the artery puncture with incidence in a total of 57 (4.57%) patients. Among subacute complications the most represented was the malposition of the placed catheter, occurring in 15 (1.2%) patients.

Infection, as the most serious chronic complication, was registered in 36 (2.6%) patients, with 1.76% incidence of gram-negative infections, most frequently caused by Kl.pneumoniae (16).

Infections in non-tunneled CVC occur by migration of microorganisms from the skin, from the spot of the catheter insertion along the catheter surface. There is also a possibility of hematogenous infection which attacks the top of the catheter or contamination of the catheter interior.

Preliminary factors for infections associated with the CVC are the patient's age - under one year of age, neutropenia, immunosuppressive therapy, the skin integrity damage - burns, gravity of initial illness, presence of a distant infection. Other risk factors involve: prolonged catheterization, frequent catheter manipulation, inappropriate aseptic technique during the catheter placement and maintenance, number of lumens and location of the catheter (17).

CONCLUSION

Central venous access is an irreplaceable invasive procedure in the intensive care units which enables quick administration of medications and liquids during cardiac arrest, saving the patient's life; administration of partial and full parenteral nutrition, invasive monitoring of cardiocircular functions, frequent blood taking for laboratory tests. Infections are leading complications associated with central venous catheterization. They are significant in causing iatrogenic morbidity and mortality, which can often be prevented. Accordingly, healthcare practitioners working with intravascular catheters need to be familiar with the indications for application and accurate placing and maintenance of catheters in order to minimize the related complications.

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Comparison of postoperative complications and factors affecting the surgical treatment of inflammatory bowel diseases

Komparacija postoperativnih komplikacija i faktora koji utiču na hirurško liječenje inflamatornih bolesti crijeva

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ABSTRACT

Introduction: Crohn's disease and ulcerative colitis are inflammatory bowel diseases (IBDs), which can be treated both medically and surgically. The aim of the study was to evaluate the results of surgical treatment of IBDs and to analyse the frequency of occurrence of early postoperative complications. Materials and methods: retrospective study included 43 patients surgically treated with inflammatory bowel diseases at the Clinic for General and Abdominal Surgery of the University Clinical Center Sarajevo in the period from 1 January 2011 to 31 December 2013. Results: in the observed period 27 patients with Crohn's disease and 16 patients suffering from ulcerative colitis were treated surgically. Postoperative complications occurred in 30.2% of patients. The most common complications are infections and dehiscence that were observed in 9.3% of cases, following ileus or obstruction in 7%, and cardiovascular and respiratory complications in 2.3% of cases. Slightly higher occurrence of complications associated with older age, male, comorbidity and urgently operated patients was confirmed, but statistically irrelevant (p>0.05). The average length of hospitalization was 10.26 ± 4.69 days. Conclusion: postoperative complications were verified in about 1/3 of surgically treated diseases with IBDs, without any significant differences between patients with Crohn's disease and ulcerative colitis. Risk factors for the occurrence of postoperative complications in these patients included persons above 50 years of age, male, presence of comorbidities and emergency patients.

Key words: inflammatory bowel diseases, postoperative complication

SAŽETAK

Uvod: Crohnova bolest i ulcerozni kolitis predstavljaju inflamatorne bolesti crijeva (IBC), koje se mogu liječiti medikamentozno i hirurški. Cilj rada je bio evaluirati rezultate hirurškog liječenja inflamatornih bolesti crijeva i analizirati učestalost javljanja ranih postoperativnih komplikacija. Materijal i metode: retrospektivna studija obuhvatila je 43 hirurški tretirana slučaja sa inflamatorrnim crijevnim bolestima na Klinici za opštu i abdominalnu hirurgiju KCUS-u u periodu od 1. januara 2011. godine do 31. decembra 2013 godine. Rezultati: u posmatranom periodu hirurški je tretirano 27 pacijenata sa Crohnovom bolesti, a 16 sa ulceroznim kolitisom. Postoperativne komplikacije su se javile kod 30,2% pacijenata. Najčešće komplikacije su infektivne komplikacije i dehiscencija u 9,3% slučajeva, te ileus ili opstrukcija u 7% slučajeva. Potvrđena je nešto veća učestalost pojave postoperativnih komplikacija udruženih sa starijom životnom dobi, muškim spolom, komorbiditetom i hitno operisanim, ali bez statistički signifikantne razlike (p>0,05). Prosječna dužina hospitalizacije pacijenata je bila 10,26± 4,69 dana. Zaključak: postoperativne komplikacije su verificirane u oko 1/3 hirurški tretiranih bolesti sa inflamatornim bolestima crijeva, bez signifikantne razlike između pacijenata oboljelih od Chronove bolesti i ulceroznog kolitisa. Riziko faktori za pojavu postoperativnih komplikacija u bolesnika sa inflamatornim bolestima crijeva su bili životna dob iznad 50 godina života, muški spol, prisutnost komorbiditeta i hitno operisanih.

Ključne riječi: inflamatorne crijevne bolesti, postoperativne komplikacije

INTRODUCTION

Inflammatory bowel diseases (IBDs) are idiopathic inflammatory, chronic diseases of the gastrointestinal tract, unpredictable in its course and that belong to the group of autoimmune diseases. The name - inflammatory bowel disease (IBD) refers to ulcerative colitis (UC) and Crohn's disease (CD), two closely related draining bowel diseases with similar clinical, pathological and epidemiological characteristics (1).

Ulcerative colitis (UC) is an inflammatory process of the rectum and colon mucosa. It is characterized by inflammation that begins in the rectum and continuously extends proximally. Intermittent exacerbations are typical, and the symptoms are characterized by bloody diarrhea associated with frequent and sudden stools and tenesmines. Disease activity varies from complete remission of the disease to fulminant with systemic toxic effects (2). Although medical therapy may reduce inflammation and regulate mayority of symptoms, this therapy does not provide a definitive treatment for this disease. Proctocolectomy or complete removal of the colon and rectum is the only definitive treatment of this disease (3).

Crohn's disease (CD) is a chronic inflammatory disease of unknown cause that can involve any part of the gastrointestinal tract. Inflammation can affect the thicknening of the entire bowel wall, and

most patients will eventually develop strictures or penetrating complications. Accordingly, the majority of patients will require surgery as part of their treatment. The probability of surgical resection within 15 years after diagnosis is 70%. Crohn's disease can be treated, but cannot be cured by surgery. The primary goal of surgical treatment is to alleviate the serious complications and achieve a better quality of life, and to preserve as much of the bowel possible (4).

The aim of this study was to analyse the incidence of early postoperative complications in patients with IBDs treated surgically at the Clinic for General and Abdominal Surgery of the University Clinical Centre Sarajevo and their association with the monitored variables.

MATERIALS AND METHODS

Retrospective descriptive study included 43 patients with diagnosed of inflammatory bowel diseases (Crohn's disease or ulcerative colitis), who were admitted and surgically treated in the period from 1 January 2013 to 31 December 2015 at the Clinic for General and Abdominal Surgery of the University Clinic Centre of Sarajevo. Operating protocols and disease histories were used as data source. The analysis covered following variables: gender, patient's age at the time of surgery, the diagnosis of disease, co-morbidity, operative indication, surgery emergency, surgical procedure, existence of postoperative complications and length of hospital stay.

Statistical analysis

Data was presented as mean ± SEM. Categorical variables were shown as frequencies. The difference in normally distributed data was tested by t-test for small samples. The analysis of categorical variables was performed using Yates' chi-square test and one-way analysis of variance (ANOVA). A p value < 0.05 was considered statistically significant for all comparisons. The analysis was performed using the statistical package IBM SPSS Statistics v19.0 (Chicago, Illinois USA).

RESULTS

In the monitored period, 43 patients with inflammatory bowel diseases were surgically treated, of which 27 (62.8) were with Crohn's disease and 16 (37.2%) with ulcerative colitis. Most of the patients were less than 50 years old, majority of which were males. Postoperative complications occurred in 30.2% of surgically treated patients. A slightly higher proportion of complications occurred in patients with ulcerative colitis, than in patients with Crohn's disease (Table 1).

Table 1 Characteristics of the study population.

	Crohn's disease	Ulcerative colitis	p
Number (%) of patients	27 (62.8%)	16 (37.2%)	<0.001
Age (years)	45,3±9,1	39,8±8,7	0,65
Male	14	11	0,47
complications (Yes)	7 (25,9%)	6 (37,5)	0,322

Data are presented as absolute numbers with percentages or as mean ± SD.

Among the operative indications, most commonly observed are acute abdomen and obstructive ileus in the 7 (16.3%) cases, followed by an abscess in 5 (11.6%) cases, and the most uncommonly fistula and diffuse intestinal polyps in the remaining two cases (Table 2).

Table 2 Operative indications.

Indications		
	N	%
Abscess	5	11.6
Perianal fistula	3	7.0
Enterocolic fistula	2	4.7
Toxic colon	3	7.0
Acute abdomen	7	16.3
Resistance to conservative therapy	4	9.3
Ileus obturaciones	7	16.3
Rectovaginal fistula	1	2.3
Enterocutane fistula	4	9.3
Gastrointestinal haemorrhage	4	9.3
Adhesive ileus	2	4.7
Diffuse intestinal polyposis	1	2.3
Total	43	100.0

The most common surgery performed was the resection of the ileum in 10 (23.3%) cases, followed by proctocolectomy in 7 (16.3%) cases, subtotal colostomy in 6 (14%) cases, right hemicolectomy in 5 (11.6%) cases, while other surgeries account for less than 10% of the cases (Table 3).

Table 3 Surgical procedures conducted in patients with IBDs.

Surgical method		
	N	%
Subtotal colostomy	6	14.0
Total colostomy	1	2.3
Proctocolectomy	7	16.3
Resection of fistula	4	9.3
Incision abscess	2	4.7
Resection of the ileum	10	23.3
Right hemicolectomy	5	11.6
Left hemicolectomy	2	4.7
Partial ileocecal resection	2	4.7
Resection of the jejunum	3	7.0
Resection of the sigmoid part of the colon	1	2.3
Total	43	100.0

The most common postoperative complications were infection and wound dehiscence, followed by ileus, cardiovascular and respiratory complications (Figure 1).

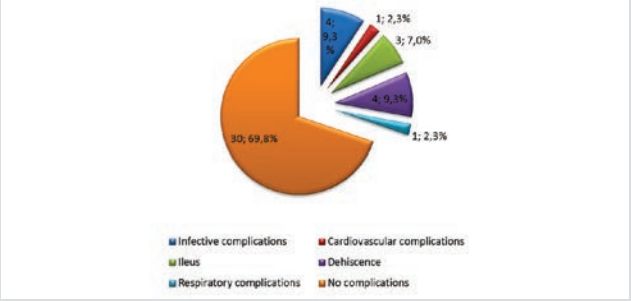


Figure 1 Frequency of occurrence of certain complications in the postoperative period.

Analysis of the incidence of complications in relation to gender, age, comorbidity and urgent surgical intervention showed a slightly higher rate of complications among males, older patients, patients with comorbidity and urgently operated patients, however, without any statistically significant difference (p> 0.05) (Table 4).

Table 4 Incidence of postoperative complications in relation to the monitored variables.

	Postoperative complications		
Variable	with	without	p
Number (%) of patients	13 (30.3%)	30 (69.8%)	<0.001
Age < 50 / >50 years	7 /6	22/8	0,184
Men / women	9 / 4	19 / 11	0,496
Comorbidities (Yes/No)	7 / 6	12 / 18	0,322
Emergency / elective surgery	8 / 5	15 / 15	0,360
Length of hospitalization	14,15±5,94	8,57±2,73	<0,001

The mean postoperative hospital stay of patients with IBDs is 10.26 ± 4.69 days. Statistically, Patients with postoperative complications were hospitalised significantly longer than the patients without complications (14.2 ± 5.9 days (range 7-25 days) vs. 8.6 ± 2.3 days (range 5-14 days; p <0.001).

DISCUSSION

Out of 43 patients who were operated in the period from 1 January 2011 to 31 December 2013, 62.8% had Crohn's disease, and 37.2% had ulcerative colitis. According to Carter and associates, 80% of patients with Crohn's have a need for surgical treatment throughout life compared to approximately 20% of patients with ulcerative colitis (5).

Postoperative complications were recorded in 30.2% of treated patients with IBD. Although this is a study with a relatively small number of patients, these data are consistent with the data from the relevant literature and other studies that suggest that the risk of surgical postoperative complication in these patients ranges from 13% to 55%, depending on the type of IBD and surgical approach (6,7,8).

The most common complications in the early postoperative period in our study were infection (9.3%) and wound dehiscence (9.3%). 69.8% of patients had no postoperative complication. In large studies of Jan and associates that included 252 patients, 32 (3.5%) had dehiscence or fistula, 74 (8.1%) had infectious complications, 2 (0.2%) had urinary tract infection, 161 (17.6%), ileus or bowel obstruction, 31 (3.4%) had cardiovascular complications, and 24 (2.6%) with respiratory complication (9).

The most common indication in our study was acute abdomen and ileus (16,3% of cases), followed by an abscess in 11.6% of cases, enterocutane fistula (9.3%) and gastrointestinal haemorrhage (9.3% of cases) , perianal fistulas (7%) and toxic colon (7%), enterocolic fistula (4.7%), adhesive ileus (4.7%), rectovaginal fistula (2.3%) and diffuse intestinal polyposis (2.3%). Similar indications for surgery may be found in literature. According to the study of Sica and associates, bowel resection is indicated in patients who are refractory to conservative therapy, in patients with obstruction, abdominal abscess, perforation, massive bleeding or even a tumour (10). Vendrell and associates include heavy bleeding, perforation and cancer in the absolute indication for colectomy in patients with UC; and the indications for such surgery are also fulminant colitis, toxic megacolon, as well as patients with dysplasia and patients resistant to conservative therapy (11).

Increased postoperative complications were observed in older patients but with no statistically significant difference (p = 0.184). Older people generally have poorer health condition and a range of co-morbidities, which, according to some studies also contribute to increased risk for developing postoperative complications. A higher incidence of postoperative complications in elderly people confirms study of Kaplan and associates, in which complications are significantly more frequent in male patients, urgently operated patients, and in patients with multiple preoperative comorbidity (12). Similar results were reported by other authors (13,14).

The analysis of the data obtained revealed that the postoperative complications were more frequent in males (at 32.1%) compared to 26.7% in females, but without statistical significance (p=0.496). There were no studies that have investigated the relationship between gender and frequency of postoperative complications.

According to the results of our study, postoperative complications in patients with IBD occurred more frequently in patients operated due to complications of ulcerative colitis (37.5%) than in patients operated due complications of Crohn's disease (25.9%), also without statistical significance (p=0.360). The study of Colombel and associates, postoperative complications occurred in 26% of surgically treated patients with Crohn's disease (sepsis in 19% of cases and unsceptical in 7% of cases (15).

Patients with IBD are typically of poorer health and often receiving immunosuppressive therapy and corticosteroids. Although surgically treated patients with IBD are mostly younger and therefore should have less comorbidity, these patients are at higher risk of developing complications, such as thromboembolic disease (12). According to the results of the study, 44.2% of patients had some comorbidity. The most common comorbidities were cardiovascular disease occurring in 20.9% of cases.

In the study of Ingram et al. that examines the risk of postoperative morbidity among patients whose surgery was planned in advance, and those who underwent emergency surgery, it was observed that patients who were operated immediately had higher risk of morbidity or death compared to patients who underwent non-emergency surgery (16). Similar observations were made in our study. Specifically, in total of 23 operated patients, postoperative complications were observed in 34.8% of patients who underwent urgent operations, versus 25% of patients who were operated electively. This distinction based on Fichers exact test was not statistically significant (p=0.360).

The most common surgery was resection of the ileum in 23.3% of cases, followed by proctocolectomy in 16.3% of cases, subtotal colectomy in 14% of cases, right hemicolectomy in 11.6% of cases, and other surgical procedures in less than 10% cases. Ileal resection was reserved for patients with Crohn's disease, which was most common in this study (62.8%). For patients with UC who need surgery, proctocolectomy was operation of choice and definitive cure for this disease. Partial colectomy should be avoided to the possible reoccurrence of the disease in the rest of the column (10).

CONCLUSION

Post-operative complications in the observed three-year period were registered in 30.2% of surgical patients treated for IBD. The incidence of complications was not statistically different between patients diagnosed with Crohn's disease and those with ulcerative colitis, and

the risk factors for the occurrence of postoperative complications in patients with inflammatory bowel disease were above 50 years of age, male gender, presence of comorbidities and emergency operated patients. The most common complications were infection and wound dehiscence, but without statistical significance. Statistically, patients with postoperative complications were hospitalised significantly longer compared with patients without complications.

Conflict of interest: none declared.

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Early MRI findings following perinatal asphyxia

Rani MRI nalaz kod perinatalne asfiksije

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ABSTRACT

Aim: to identify patterns of hypoxic-ischemic brain injury by magnetic resonance imaging (MRI) in the first 10 days of life in affected neonates. Material and methods: standard T1- and T2-weighted MRI sequences performed in the first 10 days of life in 25 asphyxiated term neonates were reviewed retrospectively. All neonates were subjected to therapeutic hypothermia. Results: out of the 25 asphyxiated neonates who underwent MRI, 18 (72%) had brain injury detected on MRI. In moderate hypoxic-ischemic encephalopathy (HIE) group 3 out of 9 (33.3%) and in severe HIE group 15 out of 16 (93.75%) showed abnormalities in their scans. The most frequent pattern of hypoxic-ischemic injury in newborns with moderate HIE was the primarly deep gray matter involvment found in 2 (66.67%) newborns, whereas in severe HIE group, it was the mixed injury pattern found in 9 (60%) neonates. Primarily cortical involvment was only present in two neonates (13.3%), both with severe HIE. One (6.67%) neonate with severe encephalopathy showed multiple punctate lesions in periventricular white matter. Conclusion: MRI can show brain damage in asphyxiated neonates during the first 10 days of life.

Key words: perinatal asphyxia, hypoxic-ischaemic brain injury, brain MRI

SAŽETAK

Cilj: identificirati obrazce hipoksično-ishemičnog oštećenja mozga na magnetnoj rezonanci (MRI) kod zahvaćene novorođenčadi. Materijali i metode: retrospektivno su analizirane standarne T1- i T2- MRI sekvence učinjene u prvih 10 dana života kod 25 asfiktičnih terminskih novorođenčadi. Sva novorođenčad su bila podvrgnuta terapijskoj hipotermiji. Rezultati: od 25 asfiktičnih novorođenčadi podvrgnutih MRI, 18 (72%) su imala ozljedu mozga detektovanu na MRI. U grupi srednje teške hipoksično-ishemijske encefalopatije (HIE), 3 od 9 (33,3%), a u grupi teške HIE 15 od 16 novorođenčadi (93,75%) je imalo abnormalnosti na skenovima. Najčešći obrazac hipoksično-ishemijske ozljede kod novorođenčadi sa srednje teškom HIE bila je primarna ozljeda duboke sive mase kod 2 (66,67%) novorođenčadi, dok je u grupi novorođenčadi sa teškom HIE, najčešći bio obrazac miješane ozljede kod 9 (60%) novorođenčadi. Primarna kortikalna ozljeda je bila prisutna kod 2 (13,3%) novorođenčeta. Zaključak: MRI može pokazati oštećenje mozga kod asfiktične novorođenčadi već u prvih deset dana života.

Ključne riječi: perinatalna asfiksija, hipoksično-ishemijska ozljeda mozga, MRI mozga

INTRODUCTION

Magnetic resonance imaging (MRI) has a large impact on the evaluation of the asphyxiated neonate. MRI can show parenchymal and intraventricular hemorrhage (1,2,3), periventricular white matter damage (1, 4,5,6), and specific patterns of brain damage (7,8,9) in affected neonates. Most studies, however, have relied on images obtained when infants were in the intermediate or late phase of injury, weeks to months after the asphyxic episode. Although imaging studies performed in the intermediate or late phase allow identification of injury patterns and, to some extent, prognostication of long-term outcome, they are temporarilly far away from the important decisions concerning patient's care. The purpose of this study was to evaluate MRI studies performed during the first 10 days of life of asphyxiated neonates in order to determine whether MRI is sensitive to hypoxic-ischemic brain injury and whether it can show specific patterns of injury during the early post injury period.

MATERIALS AND METHODS

We retrospectively reviewed MRI studies of 25 asphyxiated neonates undertaken in the first 10 days of life. All neonates were subjected to therapeutic hypothermia after fulfilling inclusion criteria of Bristol protocol (Table 1).

Table 1 Inclusion criteria for therapeutic hypothermia.

Gestational age (gestational weeks, GW)	≥ 36
Metabolic A	pH < 7 OR Base deficit ≥ 16 mmol/L in the first hour OR Apgar ≤ 5 @ 10 min OR 10 min continued resuscitation
Neurology B	Lethargy, stupor, or coma AND One or more of below • Hypotonia • Abnormal reflexes : oculomotor / pupillary • Suck: weak / absent • Clinical seizures
aEEG C	Abnormal aEEG

Data were analyzed using SPSS version 20 (SPSS, IL, USA). Clinical variables were compared using the appropriate test (X2, Fisher exact test for categorical variables, and Mann-Whitney or Kruskal-Wallis test for continuous data). Relationship between MR scores in the first and second scans was assessed using Spearman's rank correlation coeffi-

cient (ρ). Differences with p-level <0.05 were considered statistically significant.

RESULTS

General perinatal data are presented in Table 2. There were no significant differences in the perinatal data between the group with moderate and severe hypoxic-ishaemic encephalopathy (HIE) except for the need for more advance resuscitation in infants with severe HIE. All neonates with moderate HIE survived, whereas 5 (31,25%) with severe HIE died during the neonatal period.

Table 2 Perinatal characteristics of HIE newborns treated with hypothermia.

	Moderate HIE (n=9)	Severe HIE (n=16)	p
Gestational weeks ± SD	39,3 ± 1,5	38,4 ± 2,2	0,179
Birth weight (g) media ± DE	3108 ± 594	2981 ± 583	0,512
Male infants N (%)	6 (66.7)	10 (62.5)	0.746
pH mean ± SD	6,90 ± 0,13	6,90 ± 0,21	0,940
Base deficit mean ± SD	20,22 ± 5,34	20,41 ± 5,87	0,922
Lactate mean ± SD	13,82 ± 5,53	16,41 ± 5,45	0,277
1 - min Apgar score media (range)	1 (0-4)	2 (0-6)	0,638
5 - min Apgar score media (range)	4 (2-9)	4 (0-8)	0,099
10 - min Apgar score media (range)	6 (3-9)	7 (0-10)	0,624
Advance resuscitation N (%)	6 (66.67)	13 (81.25)	0.017
Neonatal death N(%)	0	5 (31.25)	0.001

Out of 25 infants who underwent MRI, 18 (72%) had brain injury detected on MRI. According to the severity of the encephalopathy, 3 out of 9 in the moderate group (33,3%) and 15 out of 16 in the severe (93,75%) showed abnormalities in their scans.

The most frequent pattern of hypoxic-ischemic injury in newborns with moderate HIE was the primary deep gray matter involvement, whereas in newborns with severe HIE, it was the mixed injury pattern. Primarily cortical involvement was only present in two neonates, both with severe HIE. One neonate with severe encephalopathy showed multiple punctate lesions in periventricular white matter (Table 3).

Table 3 MR findings in HIE newborns treated with hypothermia.

	Moderate HIE (n=9)	Severe HIE (n=16)	Toatal (n=25)
normal MRI	6 (66.67%)	1 (6.25%)	7 (28%)
abnormal MRI	3 (33.3%)	15 (93.97%)	18 (72%)
pattern of injury:			
• deep gray matter	2 (66.67%)	3 (20%)	5 (27.78%)
• cortical injury	0 (0%)	2 (13.33%)	2 (11.11%)
• periventricular white matter injury	0 (0%)	1 (6.67%)	1 (5.55%)
• mixed injury	1 (33.33%)	9 (60%)	10 (55.55%)

DISCUSSION

Finding new methods for early detection and determination of the extent of brain injury caused by asphyxia is of great importance in the treatment of asphyxiated newborns. Transfontanel sonography is the imaging study used at most institutions for the initial assessment of the asphyxiated neonate. This technique has many advantages, such as the absence of ionizing radiation and its wide availability, portability, and low cost. MRI is advanced and more powerful imaging technique compared to transfontanel sonography, but it cannot be done at the department itself. Although MRI has been established as a useful technique in the assessment of asphyxiated neonates in the intermediate (1,4,5,7) and late (8,10) stages, little has been written about the usefulness of MR imaging in the first week or 10 days of life. Improved transport conditions and experienced stuff, allow performing of brain MRI at much earlier stage even in critically ill neonates. In our study we showed that early MRI done in the first ten days of life can identify brain damage caused by asphyxia. In moderate HIE the primarily deep gray matter was involved, whereas in newborns with severe HIE, the mixed injury pattern was predominant.

CONCLUSION

MRI can show brain damage in asphyxiated neonates during the first 10 days of life. Early detection of brain injury enables early intervention which improves the final outcome. We recommend performing MRI and MRS in every full-term infant with encephalopathy and/or seizures admitted to a neonatal intensive care unit.

Conflict of interest: none declared.

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Serum total antioxidative capacity level is positively associated with cognitive functions in patients with probable Alzheimer's disease and vascular dementia

Nivo totalnog antioksidativnog kapaciteta u serumu je pozitivno povezan s kognitivnim funkcijama kod pacijenata sa mogućom Alzheimerovom i vaskularnom demencijom

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ABSTRACT

Oxidative stress (OS) has been associated with the pathogenesis of the two major types of dementia: Alzheimer's disease (AD) and vascular dementia (VD). The aim of the present study was to evaluate association between antioxidant capacity and cognitive functions in patients with probable AD and VD. This cross-sectional study included 28 patients with probable AD, 27 patients with probable VD, and 29 control subjects. Cognitive impairment was confirmed by the MMSE score. Serum total antioxidant capacity (TAC) was measured by quantitative colorimetric determination using Total antioxidant Capacity-QuantiCrom AntioxidantAssay Kit (BioAssay systems, USA; DTAC-100). Patients in AD group had significantly lower TAC level [342,67 (301,65-372,18) mM Trolox equivalents] compared to the control group [377,03 (343,78-388,40 mmol/L) mM Trolox equivalents; p=0,007], while no significant difference was found between AD and VD group [369,83 (314,82-381,75 mmol/L) mM Trolox equivalents; p=0,167] and between VD and the control group [p=0,183]. There was no significant difference in serum TAC levels between the control group and in patients with probable VD with severe and moderate cognitive impairment. Patients in AD group with severe cognitive impairment had significantly lower TAC level [336,02 (244,56-370,94) mM Trolox equivalents] compared to the control group [377,03 (343,78-388,40 mmol/L) mM Trolox equivalents; p=0,009], while no significant difference was found between patients with probable AD with severe cognitive impairment and patients with probable AD with moderate cognitive impairment [344,33 (319,39-379,35 mmol/L) mM Trolox equivalents; p=0,443] and between patients with probable AD with moderate cognitive impairment and control group [p=0,096]. Serum TAC level was significantly positively associated with MMSE score in all patients with dementia (Rho= 0,239; p=0,035), in the control group (Rho= 0,438; p=0,017) and in all patients with dementia with moderate cognitive impairment (Rho= 0,539; p=0,007). The reduced level of TAC in patients with possible AD suggests the increased level of oxidative stress in these patients, as the potential risk for the onset and progression of the disease. The measuring of the level of TAC in the serum in patients with AD could serve as a potential biomarker for diagnosis of this disease. The positive correlation between TAC and MMSE in patients with dementia, in patients with dementia with moderate cognitive impairment, as well as in the control group might suggest the protective effects of antioxidants in terms of improved cognitive function.

Key words: total antioxidant capacity, oxidative stress, Alzheimer's disease, vascular dementia, cognitive impairment

SAŽETAK

Oksidativni stres (OS) je povezan sa patogenezom dvije najčešće vrste demencije: Alzheimerovom bolesti (AD) i vaskularnom demencijom (VD). Ova studija ima za cilj da procijeni povezanost između antioksidativnog kapaciteta i kognitivnih funkcija kod pacijenata sa mogućom AD i VD. U ovoj presječnoj studiji je bilo uključeno 28 pacijenata sa vjerovatnom AD, 27 pacijenata sa vjerovatnom VD, i 29 kontrolnih ispitanika. Kognitivno oštećenje je procijenjeno MMSE testom. Ukupni antioksidativni kapacitet u serumu (TAC) je mjeren kvantitativnom kolorimetrijskom metodom pomoću Total antioxidant Capacity-QuantiCromAntioxidant Assay Kit (BioAssay systems, USA; DTAC-100). Pacijenti AD grupe imali su značajno niži nivo TAC-a [342,67

(301,65-372,18) mM Trolox ekvivalenti] u odnosu na kontrolnu grupu [377,03 (343,78-388,40 mmol / L) mm Trolox ekvivalenata; p=0,007], dok nije utvrđena značajna razlika između AD i VD grupe [369,83 (314,82-381,75 mmol / L) mm Trolox ekvivalenata; p=0.167], kao ni između VD i kontrolne grupe [p = 0.183]. Nije bilo značajne razlike u serumskom TAC-u između kontrolne grupe i kod pacijenata sa vjerovatnom VD sa teškim i umjerenim kognitivnim oštećenjem. Pacijenti AD grupe sa teškim kognitivnim oštećenjem su imali znatno niži nivo TAC-a [336.02 (244. 56-370,94) mM Trolox ekvivalenata] u odnosu na kontrolnu grupu [377,03 (343,78-388,40 mmol/L) mm Trolox ekvivalenata; p=0,009], dok nije pronađena značajna razlika između pacijenata sa vjerovatnom AD sa teškim kognitivnim oštećenjem i pacijenata sa mogućom AD sa umjerenim kognitivnim oštećenjem [344,33 (319,39-379,35 mmol / L) mm Trolox ekvivalenata; p=0.443], kao i između pacijenata sa vjerovatnom AD sa umjerenim kognitivnim oštećenjem i kontrolne grupe [p=0,096]. Serumski nivo TAC-a je značajno pozitivno povezan sa MMSE rezultatom kod svih pacijenata sa demencijom (Rho = 0.239; p = 0,035), u kontrolnoj grupi (Rho=0.438; p=0,017), kao i kod svih pacijenata sa demencijom sa umjerenim kognitivnim oštećenjem (Rho=0.539; p=0,007). Smanjen nivo TAC-a kod pacijenata sa mogućom AD ukazuje na povećan nivo oksidativnog stresa kod ovih pacijenata, kao jedan od potencijalnih rizika za nastanak i progresiju bolesti. Praćenje nivoa TAC-a u serumu pacijenata sa AD može poslužiti kao potencijalni biomarker za dijagnozu ove bolesti. Pozitivna korelacija između TAC-a i MMSE-a kod pacijenata sa demencijom, zatim demencijom sa umjerenim kognitivnim oštećenjem, kao i u kontrolnoj grupi može ukazati na protektivne efekte antioksidanasa u smislu poboljšanja kognitivnih funkcija.

Ključne riječi: totalni antioksidativni kapacitet, oksidativni stres, Alzheimerova bolest, vaskularna demencija, kognitivno oštećenje

INTRODUCTION

Oxidative stress (OS) is defined as a state in which oxidants (free radicals) overwhelm anti-oxidant defences. Overproduction of free radicals can cause oxidative damage to biomolecules (lipids, proteins, DNA), eventually leading to many chronic diseases, including neurodegenerative diseases (1,2). The antioxidant defence system composed of nonenzymatic and enzymatic antioxidants, molecules capable of stabilizing or deactivating free radicals before they attack cells (2). Total antioxidant capacity (TAC) considers all the antioxidants present in plasma and body fluids, thus providing an integrated parameter rather than the simple sum of measurable antioxidants (3). Oxidative stress is associated with ageing and several brain pathologies (4). There is growing evidence that OS has been associated with the pathogenesis of the two major types of dementia: Alzheimer's disease (AD) and vascular dementia (VD) (5). Alzheimer's disease, the most common type of dementia in older adults is an age-related progressive neurodegenerative disorder. The disease is characterized by the presence of senile plaques formed from beta amyloid (Aβ), and neurofibrillary tangles (NTFs), severe cerebral cortex atrophy and cognitive impairment (5-7). Previous studies have shown that oxidative stress in AD is manifested through high levels of oxidized proteins, advanced glycation end products, lipid peroxidation end products, formation of toxic species, such as peroxides, alcohols, aldehydes, free carbonyles, ketones, cholestenone and oxidative modifications in nuclear and mitochondrial DNA (8). Also, high levels of peripheral markers of oxidative stress and low antioxidant power have been reported in patients with AD (5). Vascular dementia is the second most prevalent form of dementia disorder caused by the damage of the blood vessels connected to the brain, caused mainly due to multiple strokes. An increased level of reactive oxygen species in the vasculature, reduced nitric oxide bioavailability, and endothelial dysfunction leading to vascular disease is associated with vascular dementia (9). High levels of oxidative stress markers and low concentration of antioxidants were observed in patients with VD, as in AD.

The present study is aimed to evaluate the association of total antioxidant capacity with cognitive functions in patients with probable Alzheimer's disease and vascular dementia.

MATERIALS AND METHODS

Subjects

We conducted a cross-sectional study which included 28 patients with probable AD, 27 patients with probable VD, and 29 community-dwelling age-matched apparently healthy, asymptomatic persons without dementia. All subjects were aged 65 years or more. The patients were institutionalized in a specialized unit of the Health Care Hospice for persons with disabilities in Sarajevo, Bosnia and Herzegovina. The clinical diagnosis was made by a senior staff neurologist and psychiatrist by NINCDS-ADRDA criteria for probable AD (10) and by NINDS-AIREN criteria for probable VD (11). Body Mass Index (BMI) for each subject was calculated as weight in kilograms divided by height in square meters. Height was measured with stadiometer and weight was measured with Toledo self-zeroing weight scale. Trained persons measured blood pressure using a mercury sphygmomanometer around the right arm after at least a 5-min rest. Subjects with a history of chronic inflammatory disease (asthma and rheumatoid arthritis), hepatic or renal insufficiency and cancer were excluded from the three groups.

Approval for the study was obtained from the local ethics committee. All procedures on human subjects were performed in the accordance with the 1975 Helsinki Declaration. Informed consent was obtained from subjects and caregivers upon careful explanation of the study procedure.

Cognitive Assessment

Global cognitive function was tested with the MMSE test (12) used for rapid screening of those with cognitive and/or intellectual deficit. It assessed orientation, short term memory, serial subtraction, constructional capacities and use of language. The total score is 30. A score of 24 was considered abnormal. All patients in the AD and VD groups had a score ≤ 20, while the score of subjects in the control group was 26-30. Patients with probable AD and VD were divided into two groups based on the total MMSE scores (13):

a) AD group with severe cognitive impairment (score 0-10)

b) AD group with moderate cognitive impairment (score 10-20)

Hachinski ischemic score

The Hachinski ischemic score (HIS) helps differentiate patients with VD from those with AD. The original scale consists of 13 items; each scale item was assigned a numeric value with double weighting applied to specific clinical features. The score 7 or more relates to vascular dementia, the 0-4 score to Alzheimer's dementia and the 4-7 score to mixed dementia (14). The score of our patients in AD group was 4 or less, and in patients with VAD the score was 7 or more.

Assessment of serum Total Antioxidant Capacity

Analysis of TAC was performed at the Department of Physiology of the Faculty of Medicine in Sarajevo. 5 mL of blood sample were taken from each subject from the antecubital vein, in the fasting state. Blood samples were centrifuged at 4°C for 10 min at 2500 rpm to obtain serum. The serum samples were stored at -80°C until the analysis. The samples were thawed at room temperature only at the time of assay. Serum total antioxidant capacity was measured by quantitative colorimetric assay, using Total antioxidant Capacity -QuantiCromAntioxidantAssay Kit (BioAssay systems, USA; DTAC-100). By this method Cu2+ is reduced by antioxidant to Cu+. The resulting Cu+ specifically forms a colored complex with a dyed reagent. The color intensity at 570 nm is proportional to TAC in the sample. Serum total antioxidant capacity was expressed in mMTrolox equivalents. Referral TAC values using this kit is within the linear range of 1.5-1000 mMTrolox equivalents.

Statistical analysis

All statistical calculations were performed with the SPSS 16 software (version 16.0, SPSS Inc, Chicago,Illinois, USA). The distribution of variables was tested by Kolmogorov-Smirnov or Shapiro-Wilk test. Values with normal distribution were expressed as mean ± standard deviation, while those without normal distribution were shown as median and interquartilerange. For continuous variables, comparison between the groups were performed by one-way analysis of variance (ANOVA) followed by multiple comparison Bonferroni test, Kruskal-Wallis test, and Mann-Whitney U-test. Additionally, Spearman correlations were used as measures of association for the continuous variables. P-values less than 0.05 were considered statistically significant.

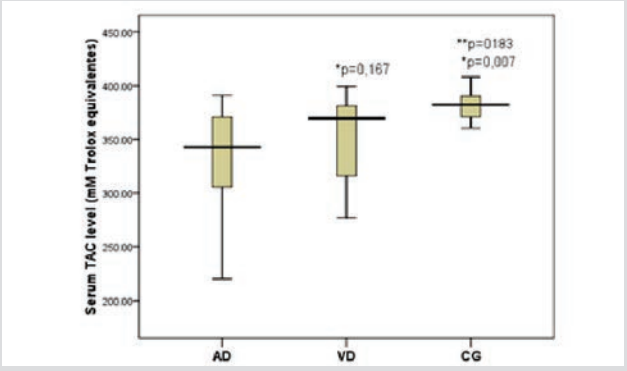
RESULTS

The characteristics of the patients and control group are shown in Table 1. There were no differences in systolic and diastolic blood pressures and BMI between the groups. The AD and VD groups had significantly lower MMSE scores than the control group (p<0.0001), but there was no difference in MMSE scores between the AD and VD groups.

Table 1 Baseline characteristics of patients in AD, VD and control group.

Variables	CG (n=30)	AD group (n=30)	VD group (n=30)
Age (years)	81.14±5.92	82.32±5.79	81.47±6.29
BMI (kg/m2)	25.18±4.25	25.11±3.69	24.89±6.00
SBD (mmHg)	132.37±22.18	134.10±21.66	132.80±20.76
DBP (mmHg)	83.16±12.46	81.09±9.82	82.19±12.28
MMSE (score)	28.25 (27.00-29.00)	9.00 (6.75-12.25) ***	12,75 (7.00-15.75) ***

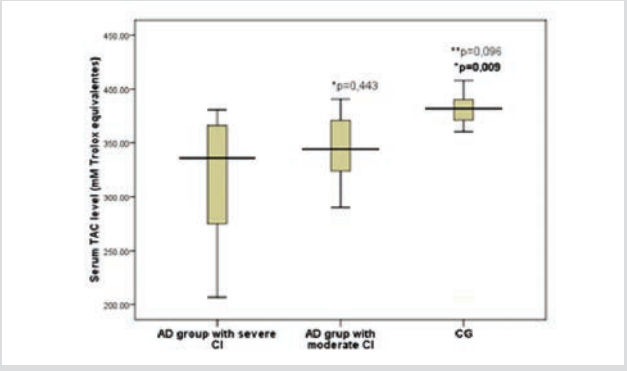
*Data as mean ± SD. and as median and interquartile range. SBP: systolic blood pressure; DBP: diastolic blood pressure: BMI: Body mass index; MMSE: Mini Mental State Examination; *** p<0.001*



*Data are reported as median and interquartile range; TAC, total antioxidant capacity; AD, Alzheimer's disease; VD, vascular dementia; CG, control group; *in comparison to Alzheimer's disease; **in comparison to vascular dementia*

Figure 1 Plasma TAC levels in patients wih Alzhemer's disease, vascular dementia and control group.

Patients with probable AD had significantly lower TAC level [342,67 (301,65-372,18) mM Trolox equivalents] compared to control group [377,03 (343,78-388,40 mmol/L) mM Trolox equivalents; p=0,007], while no significant difference was found between patients with probable AD and patients with probable VD [369,83 (314,82-381,75 mmol/L) mM Trolox equivalents; p=0,167], and between patients with probable VD and the control group [p=0,183] (Figure 1).



*Data are reported as median and interquartile range; TAC, total antioxidant capacity; AD group with severe CI, Alzheimer's disease with severe cognitive impairment; AD group with moderate CI, Alzheimer's disease with moderate cognitive impairment , CG, control group; *in comparison to AD group with severe CI, **in comparison to AD group with moderate CI*

Figure 2 Plasma TAC levels in patients with Alzheimer's disease with severe and moderate cognitive impairment and control group.

Patients with probable AD with severe cognitive impairment had significantly lower TAC level [336,02 (244,56-370,94) mM Trolox equivalents] compared to control group [377,03 (343,78-388,40 mmol/L) mM Trolox equivalents; p=0,009], while no significant difference was found between patients with probable AD with severe cognitive impairment and patients with probable AD with moderate cognitive impairment [344,33 (319,39-379,35 mmol/L) mM Trolox equivalents; p=0.443] and between patients with probable AD with moderate cognitive impairment and control group [p=0.096] (Figure 2).

There was no significant difference in serum TAC levels between control group in VD group with severe and moderate cognitive impairment (p=0.317).

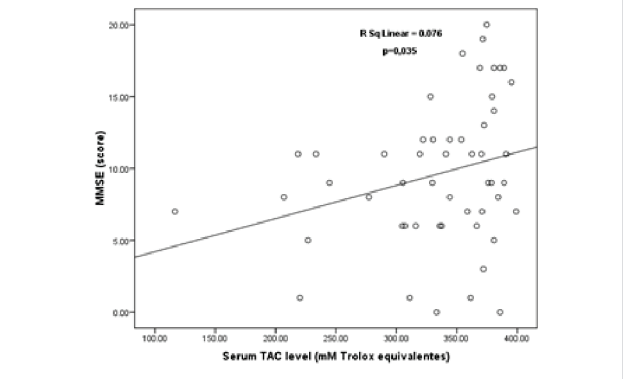


Figure 3 Association between serum TAC levels and MMSE score in all patients with probable dementia.

Serum TAC level was significantly positively associated with MMSE score in all patients with probable dementia (Rho=0.239; p=0.035) (Figure 3).

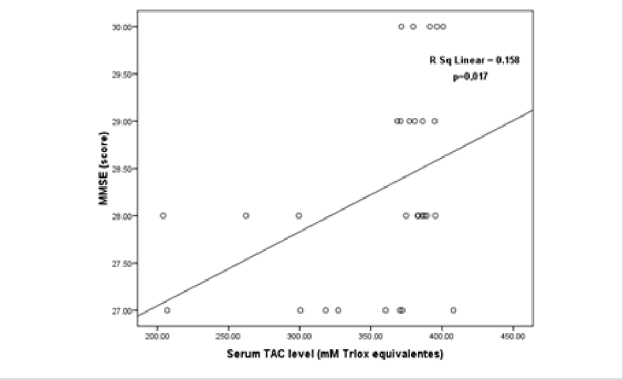


Figure 4 Association between serum TAC levels and MMSE score in the control group.

Serum TAC level was significantly positively associated with MMSE score in the control group (Rho= 0.438; p=0.017) (Figure 4).

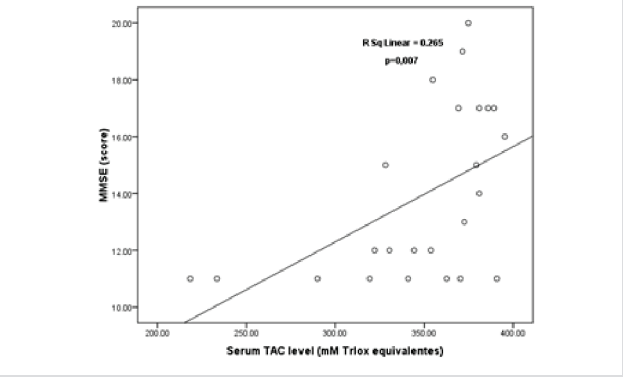


Figure 5 Association between serum TAC levels and MMSE score in all dementia patients with moderate cognitive impairment.

Serum TAC level was significantly positively associated with MMSE score in all patients with dementia with moderate cognitive impairment (Rho=0.539; p=0.007) (Figure 5).

There was no statistically significant correlation between serum TAC level and MMSE score in all dementia patients with severe cognitive impairment (Rho=0.034; p=0.864).

DISCUSSION

In our study we found that the serum levels of TAC in patients with probable AD and probable VD was lower than in healthy controls, but the only significant differences were observed between patients with probable AD and the control group. Also, no statistically significant difference was found in the serum TAC levels between AD and VD types of dementia. Our study suggests that patients with AD have lower TAC levels indicating a compromised systemic antioxidant defense which implies increased oxidative stress. These results are consistent with the findings of the previous studies (15-17) which showed lower antioxidant capacity in patients with AD compared to control subjects. According to the study of Aldred et al. (15) there was a significant decrease in plasma levels of TAC in AD patients against healthy controls. Meta-analysis by Schrag et al. (16) confirmed which TAC was decreased in patients with AD. The study by Vaisi-Raygani et al. (17) analyzed the enzymatic activities of the erythrocyte Cu-Zn superoxide dismutase (Cu-Zn SOD), glutathione peroxidase (GSH-Px), catalase (CAT), and plasma level of total antioxidant status (TAS) in AD and control groups. They showed that the level of GSH-Px and TAS activities were significantly lower in AD subjects compared to that in the control group but the Cu-Zn SOD activity was significantly higher, which support the idea that the oxidative stress plays an important role in the pathogenesis underlying AD neurodegeneration. Also, literature explains variations of antioxidant status in terms of activities and expression of antioxidant enzymes such as SOD, catalase, glutation have been observed in central nervous system as well as in peripheral tissues in patients with AD (5,18).

Regarding pathophysiology, AD patients have accumulated misfolded beta-amyloid (Aβ) a protein produced from the amyloid precursor protein (APP) and neurofibrillary tangles (aggregates of hyperphosphorilated tau protein) in brains (5). Post mortem and in vivo studies have demonstrated an accumulation of products of free radical damage in the central nervous system and in periferal tissue of subjects with AD (19). In addition to the damage caused by free radicals AD patients have high protein nitrotyrosination levels in cerebral cortex, which are associated with amyloid deposition which is closely related to oxidative stress. Along with nitrotyrosination of tau protein, synaptophysin also develops in AD brains which is consistent with a dysfunction in cholinergic synaptic transmission (20).

In our study, patients with probable AD with severe cognitive impairment had significantly lower TAC level compared to the control group, while no significant difference was found between patients with probable AD with severe cognitive impairment and patients with probable AD with moderate cognitive impairment and between patients with probable AD with moderate cognitive impairment and the control group. On the contrary, there was no significant difference in any of the TAC serum levels in patients with probable VD with severe

cognitive impairment and in patients with probable VD with moderate cognitive impairment compared to controls, as well as between these two subgroups of patients with VD.

Serum TAC level was significantly positively associated with MMSE score in all patients with probable dementia, in control group and in the subgroup of all patients with moderate cognitive impairment. However, there was no statistically significant correlation between serum TAC level and MMSE score in all patients with dementia with severe cognitive impairment. Our results are in line with recently conducted studies (Moslemnezhad et al., Zafrilla et al.).

In the study of Moslemnezhad et al. (21) plasma levels of TAC and 8-hydroxy-2'-deoxyguanosine (8-OHdG) were assesed in the patients with severe AD who were confirmed by MMSE score (0-10), CT scan and brain MRI. It showed significant decrease in TAC levels in comparison to healthy controls, and significant increase in 8-OHdG. Similarly, according to study Zafrilla et al. (22) TAC was lower in patients with severe AD compared to controls, even between the patients with moderate AD and healthy controls. Most probably in late stages of dementia all antioxidant mechanisms are exploited so that there is an unrestrainable invasion of ROS and RNS. Another fact that could be added to prevous statements is the strong positive correlation between MMSE and serum TAC levels in all patients.

According to the study of Cervellati et al. (23) the total serum nonenzymatic antioxidants were lower in late onset AD in comparison to mild cognitive impairment, which is considered as a prodromal phase of AD, and with no difference between ones with MCI which progressed to LOAD and the ones which didn't. Sekler et al. (24) did not find significant differences of ferric-reducing ability of plasma (FRAP), plasma malondialdehyde and total antioxidative capacity levels between the AD and the control group. However, when subjects exhibiting cognitive impairment were divided into subgroups according to their GDS performance score, significant differences in FRAP were found between the controls and the moderate and severe AD subgroups, and between mild and moderate/severe AD.

The brain is rich in fatty acids and is susceptible to peroxidation, consumes a lot of oxygen and doesn't have a powerful antioxidant activity. Therefore, the brain tissue in patients with neurodegenerative diseases is highly sensitive to oxidative stress (5). Superoxide generated in the mitochondrial electron transport chain is metabolized by superoxide dismutase (SOD) to form hydrogen peroxide, which should be eliminated by antioxidant enzymes. There is an evidence that extracellular, cytoplasmic and mitochondrial SOD-overexpression improve spatial memory, age-related defects in LTP and neuroprotection against drug induced toxicity respectively which could support our positive correlation among total antioxidative capacity and cognition. (20). Recent study of Baierle et al. (25) showed that oxidative stress combined with cytokines negatively correlates with cognitive performances such as attention, memory, language skills etc. The study of Sekler et al. (24) found that antioxidant capacity values measured by FRAP methods were positively correlated with MMSE. Other studies also confirmed that GR/GPx activity positively correlated with MMSE score in patients with AD (5,26). However, meta-analysis by Schrag et al. (16) found that the major contributors to antioxidant capacity of serum are urate, plasma proteins (albumin, transferrin and ceruloplasmin), and vitamin C and E (in descending order) which could open a new approach in the prevention and treatment of this disease.

CONCLUSION

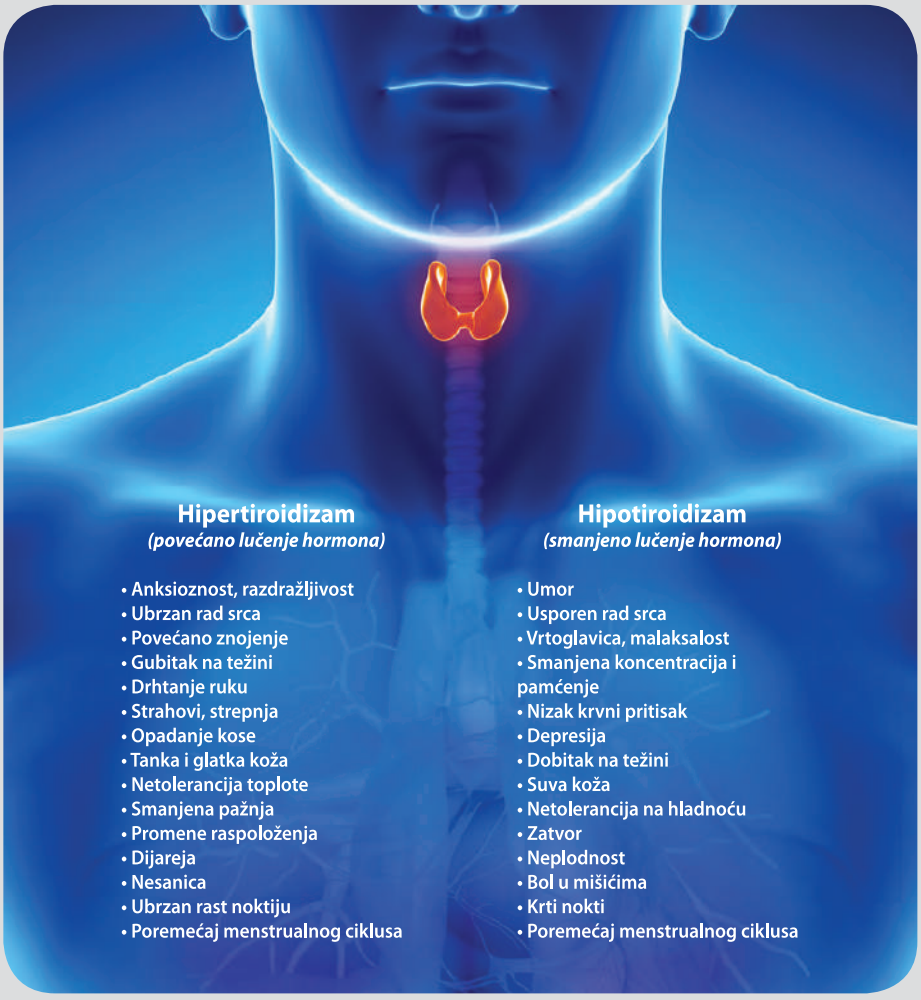
The reduced level of TAC in patients with probable AD suggests that the level of oxidative stress markers are increased in these patients and are potential risk for the onset and progression of the disease. The measuring of the TAC level in the serum of these patients could serve as a potential biomarker for the diagnosis of this disease. The positive correlation between TAC and MMSE in patients with dementia, in patients with dementia with moderate cognitive impairment, as well as in the control group might suggest the protective effects of antioxidants in terms of improved cognitive function. Our results showed that AD patients with severe cognitive impairment exhibit significant decrease in serum antioxidant protection which suggests that raising the antioxidant protection might have therapeutic benefit in patients suffering from this type of dementia.

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Myocardial perfusion scintigraphy in patients with coronary collaterals

Perfuziona scintigrafija miokarda kod pacijenata sa koronarnim kolateralama

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ABSTRACT

The aim of this study was to evaluate the effect of angiographic evidence of coronary collaterals on myocardial perfusion in patients with coronary occlusion. The retrospective study considered 50 patients with occlusion of at least one major coronary artery who underwent rest - stress myocardial perfusion scintigraphy and coronary angiography. Based on the extent of their coronary collaterals 50 patients were categorized into two groups: group of 32 patients with well-developed coronary collaterals and group of 18 patients with poor coronary collateral development. In the group with well-developed coronary collaterals we established the presence of viable myocardium in the perfusion territory of the infarct-related artery. The perfusion defects in the group with good collaterals were predominantly reversible, suggesting that coronary collaterals preserved myocardial viability in the regions subtended by a total coronary occlusion. These results suggest that the existence of well developed coronary collaterals may contribute to minimizing the infarct area and predicting the presence of viable myocardium. Excellent angiographic collaterals may prevent resting regional wall motion abnormalities but do not appear to protect against stress-induced perfusion defects.

Key words: myocardial perfusion scintigraphy, collaterals, coronary occlusion

SAŽETAK

Cilj studije je bio evaluirati uticaj angiografski evidentiranih koronarnih kolaterala na miokardnu perfuziju kod pacijenata sa koronarnom okluzijom. Retrospektivna studija je uključila 50 pacijenata sa okluzijom jedne od koronarnih arterija podvrgnutih koronarnoj angiografiji i perfuzionoj scintigrafiji miokarda u miru i opterećenju. Na osnovu angiografske razvijenosti koronarnih kolaterala 50 pacijenata je podjeljeno u dvije grupe: grupa od 32 pacijenata sa angiografski dobro razvijenim koronarnim kolateralama i grupa od 18 pacijenata sa slabo razvijenim koronarnim kolateralama. U grupi sa dobro razvijenim koronarnim kolateralama ustanovljeno je postojanje vijabilnog miokarda u perfuzionom području infarktom zahvaćene arterije. Perfuzioni defekti u grupi sa dobrim kolateralama su bili predominantno reverzibilni, što sugerira da dobre kolaterale štite miokardnu viabilnost u području sa totalnom koronarnom okluzijom. Rezultati sugeriraju da postojanje dobro razvijenih kolaterala može doprinijeti minimiziranju infarciranog područja i predikcija je postojanja viabilnog miokarda. Dobre kolaterale imaju zaštitnu ulogu i spriječavaju poremećaje regionalne miokardne kinetike u miru, ali ne spriječavaju nastanak perfuzionih defekata pri opterećenju.

Ključne riječi: perfuziona scintigrafija miokarda, kolaterale, koronarna okluzija

INTRODUCTION

The functional role of various angiographic grades for coronary collaterals remains controversial. Although myocardium supplied by well-developed collaterals may maintain normal resting perfusion, patients may still have myocardial ischemia, as manifested by anginal symptoms, ischemic electrocardiographic (ECG) changes, or abnormal myocardial perfusion studies (1,2,3). Controversy concerning the functional significance of the coronary collateral circulation in man has been compounded rather than resolved since the introduction of selective coronary arteriography despite the excellent in vivo description of collateral vessels given by several authors using this technique (4,5,6).

Myocardial imaging using thallium-201 (201Tl) is a non-invasive technique valuable in the analysis of regional myocardial perfusion. Regions of ischaemia caused by severe coronary disease appear as functional defects of tracer accumulation in the myocardial scintigram (7,8,9,10). In viable myocardium, 201 Tl is continuously exchanged between the myocardium and the blood stream, with the rate of exchange in proportion to the difference in 201Tl concentration between the myocardium and the blood stream. Therefore, after the administration of 201Tl, the myocardium concentration is greatest in areas of the myocardium with the highest blood flow. Over the time, there is a tendency toward equilibrium of 201Tl concentration in viable myocardium. The phenomenon is termed "redistribution". Therefore, a Tl 201 perfusion defect that is reversible is considered indicative of ischemic but viable myocardium (11,12,13,14).

Using ^{201}Tl imaging, we have analyzed the influence of coronary collaterals angiographically documented on myocardial perfusion in patients with coronary occlusion.

MATERIALS AND METHODS

Study population

The retrospective study included 50 patients with angiographically documented coronary artery disease. The majority of patients had complete occlusion of at least one vessel. The coronary collateral circulation was recognized in 50 patients investigated by 201Tl myocardial scintigraphy between January 2011 and May 2013.

In addition, patients with angiographically documented collaterals were assessed for the prevalence of coronary risk factors, i.e., hypertension, hyperlipidemia, smoking habits, and family history, and for the presence of diabetes mellitus.

Procedure

Coronary Angiography and Left Ventriculography

Coronary angiography was performed by the percutaneous transfemoral approach using the Judkins technique. Selective coronary angiography was performed in multiple projections.

The aim of the coronary angiography was to establish the coronary anatomy and a maximal lumen constriction of the coronary artery and the existence, origin and the direction of collaterals. On the angiograms we analyzed collaterals as adequate (average calibar greater than 1,0 mm) and inadequate (average calibar - 1mm and less). For all patients we applied left ventriculography (performed in the right anterior oblique projection) for the evaluation of its function, ejective fraction and kinetics.

Thallium – 201 Myocardial Scintigraphy

Studies of regional myocardial perfusion were performed at rest and under exertion (maximal exercise test on a bicycle ergometer). All patients had a complete physical examination, resting 12 lead electrocardiogram, and an electrocardiogram recorded during maximal exercise test on a Monark bicycle ergometer with lead V5 monitoring. During the exercise test each patients received 1,5 mCi of 201Tl through an indwelling intravenous cannula at the onset of angina pectoris or limiting dispnoea. The exercise end-point was maintained for one and a half to two minutes so that myocardial uptake of circulating tracer was maximal in the presence of symptoms. After a 10 minute recovery period, myocardial imagining was performed with subjects in the supine positions using a gamma camera and a high-sensitivity parallel-holed collimator.

Anterior, left anterior oblique 45° and 70° were routinely acquired in each patient (each position five minutes) after exercise and redistribution 3 hours later. We applied 20% symmetric energy window centered on the 72 keV peak. All projection images were stored on magneto-optic disks in a 64 x 64 matrix. The digital images were displayed on a TV monitor. After computer processing, digital scintigraphic images were described by one interpreter. Localization of perfusion defects in relation to corresponding blood vessel was determined visually in each image divided into five segments. Each defect was further classified as moderate or severe according to the degree of tracer deficit seen in that region.

The result was considered "positive" when a reversible defect was assigned to the perfusion territory of the coronary artery of interest. Defects located in the anterior wall and septal region were assigned to the left anterior descending artery, defects in the lateral wall were assigned to the left circumflex coronary artery, and defects in the inferior wall to the right coronary artery.

Statistical analysis

Statistical comparison of data was performed using the Chisquare test. Using the Chi-square test, the differences between the groups were considered significant when $p<0,001$.

RESULTS

Based on the extent of their coronary collaterals 50 patients were categorized into two groups: group of 32 patients with well-developed coronary collaterals (Figure 1) and group of 18 patients with poor coronary collateral development. In the group with well-developed coronary collaterals we established the presence of viable myocardium in the perfusion territory of the infarct-related artery and predominantly normal or hypokinetic left ventricular function. Within the group of patients with well-developed coronary collaterals ejection fraction was >50%. In the group with poor collaterals in the vast majority cases demonstrated akinesia or dyskinesia of the corresponding myocardial segment and ejection fraction was 35%

The perfusion defects in the group with good collaterals were predominantly reversible (Figure 2), suggesting that coronary collaterals preserved myocardial viability in the regions subtended by a coronary occlusion. In the group with poor coronary collateral development the perfusion defects were irreversible.

The difference between the two groups was significant at 0.001 level.

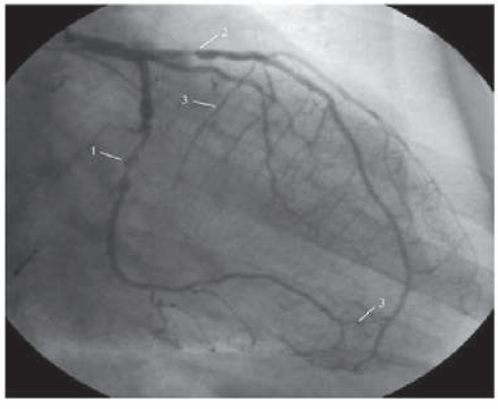


Figure 1 Coronary angiograms shows 95% subocclusion of the right coronary (1), 95% stenosis of LAD (2) and collaterals (3).

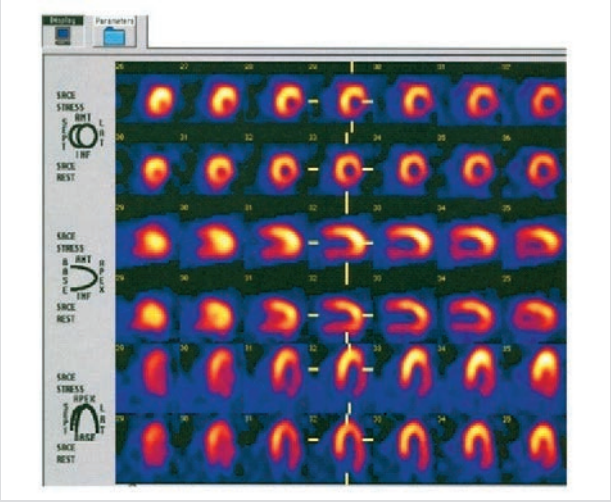


Figure 2 Analysis of perfusion scintigrams in patients with angiographically documented collaterals suggest reversible defect of antero-apical segment and ischemia (single photon emission tomography of the heart with ^{201}Tl - standard stress planar projection -45°).

DISCUSSION

The advent of coronary artery bypass graft surgery has brought renewed interest in the coronary collateral circulation and its functional significance, for the chances of successful graft patency are enhanced if the distal segment of an occluded artery has good flow maintained by collateral vessels (7).

The coronary collateral circulation in the setting of myocardial infarction has been of special interest in recent years. Well developed collaterals in a high risk population of patients

assessed at the time of acute myocardial infarction have shown to be associated with lower mortality than poor collaterals (14). Similarly, patients with good collaterals at the low risk end of the coronary artery disease spectrum, that is those with stable angina pectoris, undergoing elective PTCA are at lower risk of suffering a myocardial ischaemic event than those with poor collaterals (1,2,3,4).

The collateral development is, in general, a process of weeks to a couple of months However, the individual predisposition to have already easily recruitable interarterial connections determines the individual response to an occlusion, from severe and extensive myocardial infarction to a silent occlusion with completely preserved left ventricular function. Only well-developed collaterals will preserve viability, but on the other hand, collateral development does not depend on the presence of viability. So when we observe an occluded artery with excellent angiographic filling but akinesia in the supplied myocardial territory, we should base our indication for revascularization not on the quality of the collaterals but myocardial viability (5).

Several studies have suggested that coronary collaterals preserve resting myocardial flow but are inadequate to protect against reduced blood flow during hyperemic stress (1,7). In contrast, several studies with planar 201 Tl scintigraphy suggested that coronary collaterals also prevent stress-induced ischemia (6,8).

Fifty collateral channels were identified, graded, and compared with corresponding regions of the myocardial scintigram. In 32 patients we established angiographically well-developed coronary collaterals and the group of 18 patients with poor coronary collateral development. Tracer uptake was also graded and classified as various degrees of protection from ischaemia.

Our study showed that the perfusion defects in the group with good collaterals were predominantly reversible, suggesting that well-developed coronary collaterals preserved myocardial viability in the regions subtended by a total coronary occlusion. In the group with inadequate coronary collaterals development the perfusion defects were predominantly irreversible. In this group the absence of viable myocardium in the perfusion territory of the infarct-related artery was established.

A significant correlation between good collaterals with complete protection and poor or absent collaterals with no protection was noted. There was no difference in the protective role of homocoronary and heterocoronary collateral vessels. During the exercise the right coronary bed was preferentially protected from ischaemia, in contrast to the left anterior descending territory.

Although the mechanism of this protection is not clear, normal pressure gradient that exists between left and right ventricles may be increased during dynamic exercise, particularly during diastole in an ischaemic left ventricle, so that collateral flow is maintained or increased in the direction of the low pressure in right ventricular myocardium. This effect probably accounts for the preservation of tracer uptake in the inferior wall of the left ventricle as seen in 201Tl scintigrams (7,9,10,11,12). However, one relevant limitation of 201 Tl imaging is that relative regional myocardial ischaemia is reflected in the scintigram rather than in terms of absolute perfusion. Thus, it is possible that even though the right coronary artery territory is ischemic during exercise, it is better perfused than the left coronary artery.

CONCLUSION

These findings indicate that the presence of ischemic but viable myocardium is intimately related to the development of collateral circulation in patients with myocardial infarction, and the existence of well-developed collateral channels predicts the presence of viable myocardium in the infarct area. There was no difference in the protective role of homocoronary and heterocoronary collateral vessels. Excellent angiographic collaterals may prevent resting regional wall motion abnormalities but do not appear to protect against stress-induced perfusion defects.

Conflict of interest: none declared.

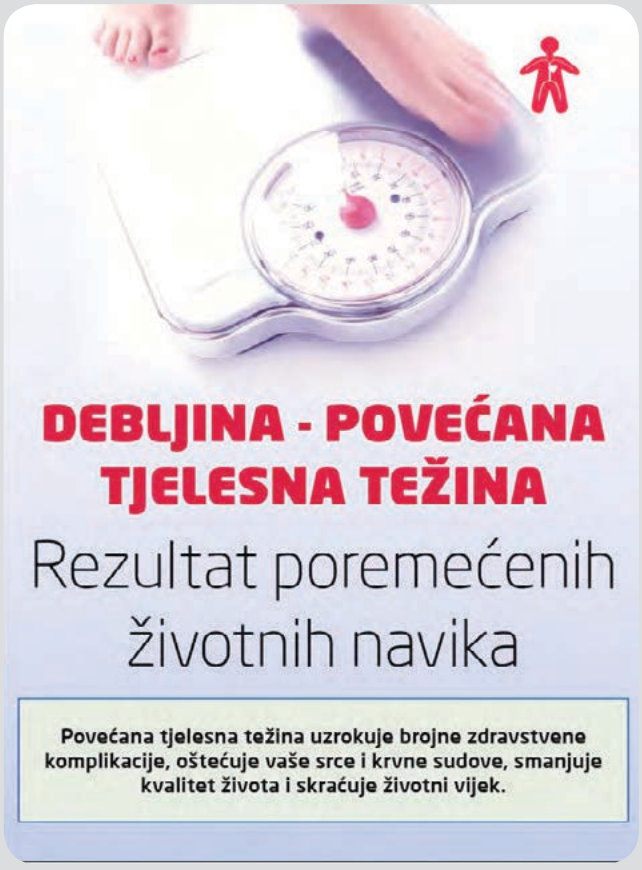
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Sensitivity and specificity of magnetic resonance imaging in breast cancerous lesions prediction in the patients with BIRADS 4 and BIRADS 5 diagnoses

Senzitivnost i specifičnost upotrebe magnetne rezonance u predikciji kanceroznih lezija dojke kod pacijentica sa postavljenom dijagnozom BIRADS 4 i BIRADS 5

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ABSTRACT

Purpose: to determine the diagnostic value of 3.0 Tesla (T) magnetic resonance imaging (MRI) in patients diagnosed with BIRADS 4 and BIRADS 5 on MRI in comparison to histological findings. Materials and methods: the study was conducted at the University Clinical Center Sarajevo in the period from September 2012 to September 2014. A total of 116 patients (mean age 52 years, range 26–81 years) classified as BIRADS 4 and BIRADS 5 were included in the study. Breast MRI findings were compared to the histopathological results. Results: malignant lesions were found in 80 patients (68.97%), whereas 36 were benign (31.03%). Malignant lesions were mostly found in patients classified as BIRADS 4C and BIRADS 5 ($p<0.001$), high percentage of them being high grade carcinomas. There were 74 true positive (63.80%), 23 false positive (19.80%), 13 true negative (11.20%), and 6 false negative breast (5.20%) MRI findings, resulting in high sensitivity, specificity, positive predictive value and negative predictive value of 92.50% (74/80), 36.11% (13/36), 76.28% (74/97) and 68.42% (13/19) respectively. Higher percentage of false negative (7.50%) and consequently lower negative predictive value (68.42%) in our study was due to dichotomy, while we considered BIRADS 4A as negative marker for cancer, and BIRADS 4B, BIRADS 4C and BIRADS 5 as positive marker. Conclusion: 3T MRI has high sensitivity and reasonable specificity in breast cancer prediction. It is the most reliable imaging method in malignancy prediction and should be used in all suspicious lesions in order to get optimal outcomes.

Key words: breast, cancer, MRI, BIRAD

SAŽETAK

Cilj: utvrditi dijagnostičku vrijednost MRI aparata 3.0 Tesla kod pacijentica dijagnosticiranih kao BIRADS 4 i BIRADS 5 prilikom MR pregleda u poređenju sa patohistološkim rezultatima. Materijali i metode: studija je sprovedena na Univerzitetskom kliničkom centru u Sarajevu u periodu od septembra 2012. do septembra 2014. godine. U studiju je uključeno 116 pacijentica (dobi 26 - 81 godine, prosječne starosti 52 godine). Nalaz MR dojki je upoređen sa histopatološkim rezultatima. Resultati: maligne lezije su nađene kod 80 pacijentica (68,97%), dok je 36 bilo benignih (31,03%). Maligne lezije su većinom nađene kod pacijentica koje u prethodno klasificirane kao BIRADS 4C i BIRADS 5 ($p<0.001$), sa visokim procentom karcinoma visokog stepena. Bilo je 74 stvarno pozitivnih (63.80%), 23 lažno pozitivnih (19.80%), 13 stvarno negativnih (11.20%) i 6 lažno negativnih (5.20%) MRI nalaza, što je rezultiralo visokom senzitivnošću, specifičnošću, pozitivnom prediktivnom vrijednosti i negativnom prediktivnom vrijednosti od 92,50% (74/80), 36,11% (13/36), 76,28% (74/97) i 68,42% (13/19). Veći procenat lažno negativnih (7.50%) i konsekutivno niža negativna prediktivna vrijednost (68.42%) u našoj studiji je nastala usljed dihotomije, jer smo BIRADS 4A smatrali negativnim markerom za kancer, dok smo BIRADS 4B, BIRADS 4C BIRADS 5 smatrali pozitivnim markerom. Zaključak: 3T MRI ima visoku senzitivnost i razumnu specifičnost u predikciji karcinoma dojke. To je najpouzdanija slikovna metoda u predikciji maligniteta i trebala bi se koristiti kod svih suspektnih lezija da bi se dobio optimalan ishod.

Ključne riječi: dojka, karcinom, MRI, BIRADS

INTRODUCTION

Breast cancer is the most common cancer in Europe for females comprising 22.9% of all cancers in women (1,2,3). According to „Cancer Research UK" with over 464.000 new cases diagnosed in 2012 and the lifetime risk of developing breast cancer is 1 in 8 for women in UK (2010). It is the leading cancer killer among women aged 20-59 years in high-income countries (2,3). In Europe the highest world age-standardized incidence rates for breast cancer are in Belgium, and the lowest are in Bosnia Herzegovina. However, the data of Institute for Statistics of the Federation of BiH show that breast cancer is the most common cancer in women in Bosnia and Herzegovina, with the highest incidence among all carcinoma in women (25.0%) in 2010 and the highest morbidity rate (52/100 000), which is similar to other countries.

Contrast-enhanced (CE) magnetic resonance imaging (MRI) of the breast is now the main radiological method for the detection and char-

acterization of breast lesions (4). MRI has shown superior potential for quantification of tumor size and follow-up after chemotherapy (5).

Several studies assessed the overall diagnostic accuracy of breast MRI in patients with suspicious breast lesions, reporting a high sensitivity of around 90%, and a considerably lower specificity of 70-75%. Furthermore, most breast MRI studies were performed using a 1.5 Tesla (T) MRI system (4,6). Imaging at 3.0 T MRI may have higher quality, and increased diagnostic confidence, but this difference does not prove higher sensitivity and/or specificity of MR imaging at 3.0 T (7).

The purpose of this study was to determine the diagnostic value of 3.0 T MRI in patient with suspicious breast malignancy, using Breast Imaging Reporting and Data System (BIRADS) and compared to patients' histological findings.

MATERIALS AND METHODS

Study design and patients selection

The retrospective data research was conducted in accordance with the Code of Ethics of the World Medical Association of Helsinki Declaration.

In the period from September 2012 to September 2014 a total of 360 breasts MRI were acquired. Investigations were performed in order to clarify uncertain lesions or to evaluate the expansion of disease, considering involvement and metastatic changes of axillary lymph nodes.

Patients classified as BIRADS 4 and BIRADS 5 were included in the study, and 116 histopathological findings were collected. Patients with changes classified as BIRADS 1, BIRADS 2, BIRADS 3, usually did not undergo biopsy, but rather follow up, BIRADS 6 was already known cancer and BIRADS 0 was excluded from the study as it could not be sorted as malignant or benign lesion and needed additional investigations.

Mean age of the examined women was 52 years (range 26–81 years, standard deviation (SD ±11.7). Mean age of 80 patients with histopathologically proven breast carcinoma was 54 years (range 26-81 years, standard deviation (SD ±11.5). Mean age of 36 patients with benign histopathological findings was 45 years (range 29-69 years, standard deviation (SD ±9.9).

Lesions were defined as malignant if surgery confirmed invasive or in situ carcinoma. Benign lesions were confirmed by surgical excision or by core biopsy. Histopathological diagnoses were performed by institutional pathologists.

Magnetic resonance imaging

All studies were performed using a 3-T clinical MR system (Siemens Medical Systems, Erlangen, Germany) with double breast coil and patients placed in prone position. Three plane localizer was followed by 3 precontrast sequences: Turbo inversion recovery magnitude sequence (TIRM) transversal (slice thickness 3 mm), T2 diffusion short-tau inversion recovery (STIR) sequences (slice thickness 4 mm) and T1 flash 3d (slice thickness 0,9 mm).

The dynamic study with contrast was made in T1 flash 3D sequence (slice thickness 0,9 mm) followed by seven consecutive measurements. Gadovist contrast medium was used (1.0 mmol/l, 0.1 mml/kg body weight).

All lesions were analyzed, the signal-intensity (SI) time curve was made in all lesions and in conclusion BIRADS classification was done.

Data analysis

Statistical analyses were performed with statistical software (SPSS Statistics 20) Microsoft Excel 2010. Descriptive statistical methods were used, and chi-square test for comparing categorical variables. The level of statistical significance was set as $P<0.05$.

RESULTS

Histopathological results

Out of the total of 116 patients classified as BIRADS 4 or BIRADS 5, 80 patients (68,97%) had malignant and 36 benign (31,03%) lesions (Table 1).

Table 1 Histopathological distribution of lesions classified as BIRADS 4 or BIRADS 5.

Malignant lesions	Frequency (f)	Percentage (%)	Benign lesions	Frequency (f)	Percentage (%)
Invasive ductal carcinoma	59	35.5	Fibrocystic breast disease	27	16.3
Invasive lobular carcinoma	14	8.4	Fibroadenoma	16	9.6
Ductal carcinoma in situ	15	9	Ductal hyperplasia	7	4.2
Lobular carcinoma in situ	10	6	Papilloma	5	3
Ductal carcinoma lymph nodes metastasis	3	1.8	Sclerosing adenosis 3	3	1.8
Papillary carcinoma	1	0.6	Mastitis	2	1.2
			Radial scar	2	1.2
			Adenoma	1	0.6
			Intramammary lymph nodes	1	0.6

Two patients (2.90%) had bilateral carcinoma and in both cases histopathological type was invasive ductal carcinoma.

Nineteen patients (27.5%) had two or more malignant lesions, mostly invasive carcinoma with in situ component, while invasive ductal carcinoma and invasive lobular carcinoma at the same side was found in four patients (5.80%). Fourteen patients (51.9%) had two or more benign lesions present in one breast.

Metastatic changes of axillary lymph nodes were found in 36 patients with breast carcinoma (45.0%), mostly in patients with invasive ductal carcinoma (86.1%) or with invasive lobular carcinoma (13.9%).

Cancer grade 3 was found in 29 patients (43.3%) as well as grade 2, while only nine patients (13.4%) had cancer grade 1.

BIRADS classification using MRI

BIRADS classification in MRI was done in accordance with the guideline of the Agency for Health Care Research and Quality in the United States (AHRQ) and the guideline of the National Breast Cancer Consultation in the Netherlands (NABON), which state that breast lesions classified as BIRADS 1 and 2 require no further diagnos-

tic work-up or follow-up other than routinely required. The work-up for a BIRADS 4 or BIRADS 5 lesion demands a biopsy procedure, because the chance of malignancy for BIRADS 4 lesion varies from 2 to 95% and for BIRADS 5 lesion higher than 95%. The probability of a BIRADS 3 lesion to be malignant is considered to be less than 2%. Therefore, the work-up of BIRADS 3 lesion can be a biopsy or follow-up mammography after 6 months (Table 2) (8,9).

Table 2 BIRADS classification.

BIRADS CATEGORIES	DESCRIPTION	FURTHER ACTIONS
BIRADS 0	Incomplete: Need additional imaging evaluation	Finding for which additional evaluation is needed
BIRADS 1	Negative	No abnormal enhancement, no lesion found (routine follow-up)
BIRADS 2	Benign finding	Benign, no malignant features (routine follow-up)
BIRADS 3	Probably benign finding	Probably benign finding (short interval follow-up)
BIRADS 4*	Suspicious abnormality	Low to moderate suspicion for malignancy (biopsy should be considered)
BIRADS 5	Highly suggestive of malignancy	High probability of malignancy (appropriate action should be taken)
BIRADS 6	Known cancer	Biopsy-proven malignancy diagnosis on the imaged finding prior to definitive therapy (appropriate action should be taken)

**Note: 4A: low suspicion for malignancy; 4B: intermediate suspicion of malignancy; 4C: moderate concern, but not classic for malignancy*

In patients classified as BIRADS 4A and BIRADS 4B in MRI mostly benign lesions were found in histopathological analysis. In patients classified as BIRADS 4C, and especially in those classified as BIRADS 5 mostly malignant lesions were found (Figure 1). The difference was statistically significant (Chi-Quadrat-Test, p<0.001).

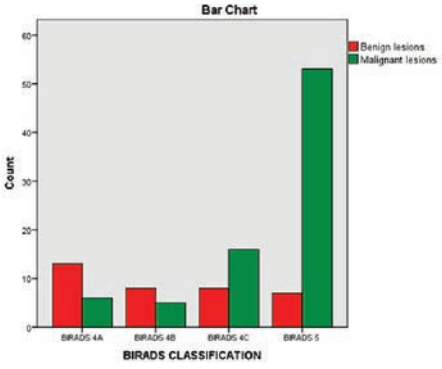


Figure 1 Patohystologically proven benign and malignant lesions in patients previously classified as BIRADS 4A, BIRADS 4B, BIRADS 4C or BIRADS 5.

Malignant lesions were mostly found in patients classified as BIRADS 5. The difference is statistically significant (p<0.001).

Patients with cancer grade 1 were mostly classified as BIRADS 4C in MRI, while patients with cancer grade 2 or 3 were previously mostly classified as BIRADS 5 (Figure 2). The difference is statistically significant (Chi-Quadrat-Test, p=0.001).

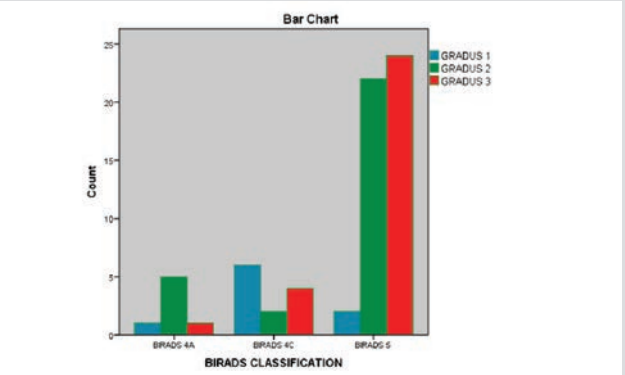


Figure 2 Gradus 3 was mostly found in patients classified as BIRADS 5, as well as gradus 2. The difference was statistically significant (p=0.001).

There were 74 true positive (63.80%), 23 false positive (19.80%), 13 true negative (11.20%) and 6 false negative (5.20%) breast MRI findings. Among malignant lesions 92.50% were true positive (classified as BIRADS 4B/4C/5 in MRI) and 7.50% were false negative (classified as BIRADS 4A in MRI), while among benign lesions 35.10% were true negative (classified as BIRADS 4A in MRI) and 63.90% were false positive (classified as BIRADS 4B/4C/5 in MRI) (Table 3).

Table 3 Summary of the results showing percentage of true positive, false positive, true negative and false negative in benign vs. malignant lesions.

	True positive	False positive	True negative	False negative	Total
Benign lesions	0	23	13	0	37
	0.00%	63.90%	35.10%	0.00%	100.00%
Malignant lesions	74	0	0	6	79
	92.50%	0.00%	0.00%	7.50%	100.00%
Total	74	23	13	6	116
	63.80%	19.8%	11.20%	5.20%	100.00%

**Note: BIRADS 4A was considered as negative marker, while 4B, 4C and 5 were considered as the positive marker in order to count true positive (TP), true negative (TN), false positive (FP) and false negative (FN) lesions.*

Malignant lesions were mostly true positive and only 7.50% were false negative. Figure 3 shows hyperintense lesion, ring like postcontrast opacification, and signal-intensity (SI) time curve type 3. It was classified as BIRADS 5, hystopathological diagnosis was invasive ductal cancer.

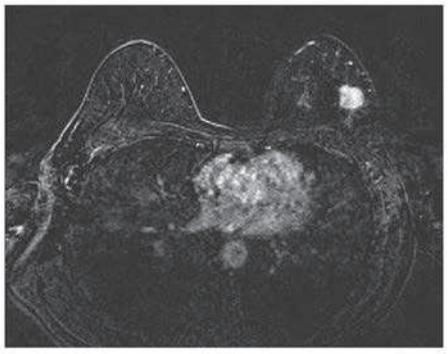


Figure 3 Invasive ductal cancer in 48 years old patient previously classified as BIRADS 5 (True positive).

False positive changes included mostly fibrocystic breast disease, fibroadenoma, and radial scar. Figure 4 shows lobulated lesion, which shows inhomogenous postcontrast opacification, the signal-intensity (SI) time curve was type 1. It was classified as BIRADS 4B, hystopathological diagnosis was radial scar.

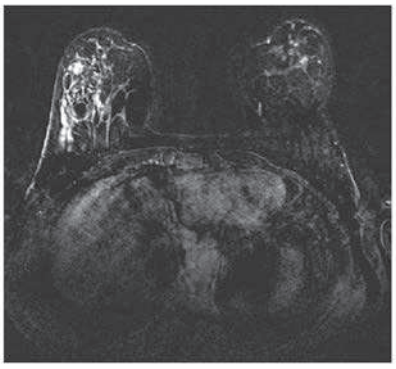


Figure 4 Radial scar in 47 years old patient previously classified as BIRADS 4B (False positive).

False negative changes included mostly ductal carcinoma in situ and lobular carcinoma in situ, but also one case of invasive ductal cancer. Figure 5 shows small lesion (9 mm) subcutaneously, in area of the previous surgical scar that shows intensive postcontrast opacification. It was classified as BIRADS 4A, hystopathological diagnosis was invasive ductal cancer.

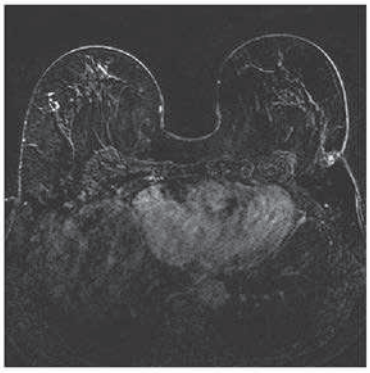


Figure 5 Invasive ductal cancer in 69 years old patient previously classified as BIRADS 4A (False negative).

Sensitivity of MRI in detecting breast carcinoma in our study was 92.50%, while specificity was 36.11%. Positive predictive value (PPV) was 76.28%, while negative predictive value (NPV) was 68.42%.

DISCUSSION

Detection of breast cancer is the most important role of breast imaging. Mammography is still the primary imaging modality in early detection of breast abnormalities and clinical trials have established that screening with mammography may decrease breast cancer mortality (10).

However, mammography has its limitations regarding both sensitivity (65.6-85.5%) and specificity (87.7–94.3%), which are dependent on age and breast density (8) and in dense breasts it is significantly decreased, down to only 30%-48%. Limited sensitivity is also noted in breasts with postsurgical scars. At the same time MRI has high sensitivity (over 90%), but still variable specificity (37%-100%) (1,11,12).

In our study we also showed high sensitivity of MRI in detecting breast carcinoma (92.50%), while specificity was 36.11%. Positive predictive value was also quite high (76.28%), but negative predictive value was lower (68.42%). False positive changes in our study included mostly fibrocystic breast disease with ductal hyperplasia and fibroadenoma, but also radial scar, papiloma and mastitis. According to Orel SG et al. the most common false positive changes include inflammatory changes and fat necrosis (13). False negative changes included mostly ductal carcinoma in situ and lobular carcinoma in situ, but also one case of invasive ductal cancer. This is in correlation with other studies (13).

Higher percentage of false negative (7.50%) and consequently lower negative predictive value (68.42%) in our study was due to dichotomy, while we considered 4A as the negative marker for cancer, because these lesions were suspected to have low suspicion for malignancy, and 4B, 4C and 5 were considered as the positive marker. We would probably get better results if we had included BIRADS 3, BIRADS 2 and BIRADS 1 in study, while these groups were more likely to be benign than BIRADS 4A.

It is well known that major limitation of mammography is differentiation of benign and malignant lesions, and approximately 75% of mammographically detected suspected lesions are proved to be benign at biopsy. The potentially limited specificity of MRI may be due to enhancement that is seen in many benign lesions, including fibroadenoma, proliferative and nonproliferative fibrocystic change, and mastitis, as well as in radial scars and atypical ductal hyperplasia. It is also difficult to differentiate enhancing of recurrent cancer from postradiation enhancing in the early period (first 18 months). Normal breast tissue can also enhance, depending of the menstrual cycle period (13).

Menzes G, et all. are pointing that value of MRI relies on detecting multifocal, multicentric or contralateral disease unrecognized on conventional assessment; recognition of in ductal carcinoma in situ; following the response to neoadjuvant chemotherapy; detecting an occult primary breast cancer in patients presenting with metastatic cancer in axillary nodes; and detection of cancer in dense breast tissue (1).

MRI findings should be correlated with mammographic findings to ensure that a lesion identified on MR images corresponds to that identified on mammograms. If suspected lesion is not proven in MRI, further recommendations must be based on mammographic, ultrasonographic and clinical findings (13). Even though there is no evidence to support MRI screening of patients treated by breast conserving surgery, Zebic-Sinkovec M, et al. suggest one whenever the findings of conventional imaging are inconclusive or it is not possible to sufficiently image the primary tumor with mammography (14).

Radiology development provides early detection. Together with improved treatment modalities it leads to longer and comfortable life and nearly 90% of women diagnosed and treated for breast cancer live more than 5 years after the initial diagnosis (15).

Due to high percentage of high grade carcinomas in our study with almost half of patients having metastatic changes of axillary lymph nodes, findings suggest that MRI examination should be done in all pa-

tients previously classified as BIRADS 4 or BIRADS 5 in mammography and ultrasonography. However, MRI in the high risk population is still not widely used. Using an abbreviated MRI protocol (fewer number of sequences) to perform high risk MRI screening may be an alternative to a full diagnostic examination. Shortening of time without sacrificing the ability to detect breast carcinoma should be cost effective and make MRI screening more available diagnostic imaging in breast cancer detection (16).

Limitations of our study related to a small number of patients. We considered 4A as the negative marker for cancer as we stated before thus getting higher percentage of false negative and consequently lower negative predictive value. Our study is also retrospective data review, and larger randomized prospective studies are necessary to be conducted, including all BIRADS categories, in order to confirm the clinical value of breast MRI.

CONCLUSION

3T MRI has high sensitivity and reasonable specify in the cancer prediction for BIRADS 4 and BIRADS 5 lesions. It is the most reliable imaging method in the malignancy prediction. MRI should be used in all suspicious lesions in order to detect disease that is occult on conventional imaging, especially in inconclusive cases, women with mammographically dense breasts, in breasts with postsurgical scars and the ones with high likelihood of breast cancer. Abbreviated MRI protocol should also be considered and could be used as adjunct to conventional imaging. Making MRI screening more available diagnostic imaging in breast cancer detection should lead to better treatment and more optimal outcomes.

Conflict of interest: none declared.

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Institute of Virology and Immunology of the Faculty of Medicine University of Sarajevo and Virology Department of the Institute of Microbiology, Parasitology and Immunology "Dr. Robert Fried", UMC Sarajevo from 1953 to 1992

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EXCERPTS FROM HISTORY OF MEDICINE

The Institute of Virology and Immunology of the Faculty of Medicine University of Sarajevo, the first institute of this kind in the former Yugoslavia, was founded on January 23, 1953. Prof. Dr. Aleksandar Terzin, physician-virologist, visionary enthusiast and scientist, was appointed the first Director of the Institute.

Establishment of the Institute at the Faculty of Medicine University of Sarajevo enabled the performance of comprehensive scientific, curricular and educational professional activities intended for doctors, scientists and students of the Faculty of Medical in Sarajevo. The Institute was located on the fourth floor of the Faculty of Medicine main building, organized in 22 premises including six laboratories, three supplementary laboratories, two teaching staff offices, one office, two laboratory glassware washrooms and several premises for experimental animals. The Institute was equipped through the funds provided by the World Health Organization (WHO), Commission for International Healthcare Cooperation (KOMNIS), Yugoslav Red Cross Central Board, Institute of Social Insurance and other organizations.

With the aim to improve the education process at the Faculty of Medicine University of Sarajevo, in January 1953 Dr. Aleksandar Terzin was appointed as Associate Professor of Virology. Danica Hlaca, BS in Biology, was appointed as assistant in the field of virology within the course of Microbiology in December 1954, whereas Dr. Maksimilijan Fornazaric was also appointed as assistant in this field in November 1955. At the same time, from six to eight students were also engaged as demonstrators within the course of Microbiology. In the upcoming period the aforementioned teacher and his associates made progress in their curricular, scientific and professional work and were promoted to the corresponding teaching ranks at the Faculty of Medicine University of Sarajevo and at other medical schools in the former Yugoslavia. Thanks to the appointment of the assistants, progress was made in the process of education, and virology practice program was expanded, all aimed at improved education of students. In 1953, Prof. Dr. Robert Fried (Director of the Institute of Microbiology) and Prof. Dr. Aleksandar Terzin had the "Manual of Standard Methods for Routine Microbiological Procedures" published by the Medical Book Belgrade – Zagreb, which was re-published in 1967. In 1955, Prof. Dr. Aleksandar Terzin published the textbook "Fundamentals of Medical Virology", Medical Book Belgrade - Zagreb, the first of this kind in the former Yugoslavia.

During the 1952-1992 aside from professional activities, significant scientific research work was performed resulting in modernization and introduction of several fundamentally important diagnostic methods in the field of virology, which were subsequently accepted and applied in other world laboratories. Thus, special attention was given to the study of causal agents of viral etiology infectious diseases which were for the first time discovered in our region or wider.

In 1958, Prof. Dr. Aleksandar Terzin was promoted to Full Professor along with the appointment of other teachers in the field of microbiology. Jovanka Vidakovic, BS in Biology, was appointed as assistant in 1960, whereas in 19961 assistants Danica Hlaca and Dr. Maksimilijan Fornazaric were promoted to the Institute lecturers in the field of virology and immunology. Dr. Maksimilijan Fornazaric worked at the Institute until 1963, when he left for Slovenia. Danica Hlaca dedicated her scientific work to the study of viral etiology of cold conditions in humans and, in 1965 she habilitated herself presenting the thesis "Contribution to knowledge about the common cold of viral etiology" before the Academic Council of the Sarajevo Faculty of Medicine.

In early 1964, Prof. Dr. Aleksandar Terzin left the Institute and the Sarajevo Faculty of Medicine, and Prof. Dr. Jakob Gaon was appointed as Acting Director of the Institute of Virology and Immunology.

Courses in the field of virology were organized for postgraduate students who since 1963 attended the course in microbiology at the Institute of Virology and Immunology, with one student defending a master thesis. During the last several years of the Institute's operations – until its merger with the Institute of Microbiology in 1967, internship in virology was organized as part of the specialization course in the field of microbiology. Accordingly, twenty-three physicians, six biologists, two

masters of pharmacy and one veterinarian completed the virology internship program. Scientific research at the Institute of Virology and Immunology was also advancing throughout these years. During the 1953-1964 period, scientists from major centers of Yugoslavia, such as Dr. Bogoljub Arsic, Dr. Bozidar Birtasevic and Matuka Stjepan, Doctor of Veterinary Medicine also joined Prof. Dr. Aleksandar Terzin, Danica Hlaca and Dr. Maksimilijan Fornazaric in their scientific curricular work at the Institute. They were studying the etiology of viral diseases, as well as little-known immunological occurrences in human organism. For the first time in Yugoslavia, influenza virus types A and B as well as some other respiratory viruses, including the causal agent of psittacosis, were isolated from the samples collected from patients at the Institute of Virology and Immunology. At the same time, serological diagnosis of diseases caused by influenza virus and viruses of atypical pneumonia, lymphocytic choriomeningitis and psittacosis were introduced at the Institute. Heat-stable complement-fixing antigens of Coxiella burnetii were also studied as part of the scientific work at the Institute. After being studied and standardized for a long time, the complement-binding reaction (CBR) was introduced in regular laboratory diagnostics and over a short period of time accepted as a standard diagnostic method in other world laboratories. Possibilities of production and stabilization of antigen components for CBR were also studied. In the field of basic scientific research, special attention was given to the study and differentiation of inhibitors and incomplete serum antibodies, as part of the study related to certain immunological occurrences in human organism. Special attention was also given to the study and diagnosis of trachoma within the process of its eradication from the area of Bosnia and Herzegovina.

During the 1953 -1964 period, Prof. Dr. Aleksandar Terzin was elected as expert of the World Health Organization (WHO) and the Institute was registered as the WHO Regional Influenza Center. In 1964, Danica Hlaca isolated certain strains of rhinovirus and for the first time in Yugoslavia performed serological tests on this group of viruses. The institute also established a laboratory for arbovirus infections analysis, which in cooperation with virologists from the Sarajevo Faculty of Veterinary for the first time in Yugoslavia managed to isolate some of the representatives of arboviruses of human health significance. Specificaly, the first clinical epidemiological tests were performed for etiology of viral diseases of the respiratory system, causal agents of aseptic viral meningitis, as well as causal agents of the cold of viral etiology. Employees of the Institute of Virology and Immunology presented the results of their successful scientific and professional researches in the country and abroad. During his work at the Institute, Prof. Dr. Aleksandar Terzin held 30 expert lectures at various meetings in the country, at virology centers in the USA and at the London School of Hygiene and Tropical Medicine in the UK.

Since the establishment of the Institute of Virology and Immunology of theSarajevo Faculty of Medicine, special attention was given to the education of teachers and associates in both professional and scientific terms, as well as to the education in other organizational forms. Based on the records, during the 1955-1956 period Prof. Dr. Aleksandar Terzin spent four months in 10 countries of the Western Europe, and in 1957 he spent two months at the London School of Hygiene and Tropical Medicine in the UK in, as part of the WHO program aimed at improvement of education in the field of virology. Prof. Dr. Terzin also spent 20 days in the USSR and in Czechoslovakia respectively, as part of scientific cooperation that the USSR Academy of Sciences and Arts and Czechoslovakia organized in 1958 and 1960, respectively. In the 1959-1960 period he attended the events organized by Rockefeller Grant at 16 different institutes in the USA during and in the 1961-1962 period, he spent 12 months at the Naval Medical Research Institute in Bethesda, USA.

In the 1960-1961 period Dr. Maksimilijan Fornazaric attended a 13-month specialization course in Newcastle, London, as part of the WHO organized program, whereas lecturer Danica Hlaca spent five months at the Central Laboratory of Virology in Stockholm. Danica Hlaca also spent six months in Birmingham, the UK, and one month in Prague and Bratislava, Czechoslovakia, and on several occasions she visited the virology laboratories in Ljubljana, Slovenia, Belgrade, Serbia and Zagreb, Croatia.

The Institute of Virology and Immunology was merged with the Institute of Microbiology to form a joint institute, the Institute of Microbiology, during the process of reorganization of the Sarajevo Faculty of Medicine in 1967. The following departments were established within the Institute: Department of Bacteriology, Department of Virology, Department of Parasitology and Mycology, Department of Culture Media, Department of Administration and the Department of Immunology.

With the aim to improve the process of education in the overall territory of Yugoslavia, the first Yugoslavian textbook "Microbiology and Parasitology" was drafted for the students of medical schools, edited by Prof. Dr. Bogdan Karakasevic and published by the Medical Book Belgrade - Zagreb in 1962. Prof. Dr. Aleksandar Terzin took part in the publication of the first and second edition of the textbook by drafting three chapters in 1969.

The Institute of Microbiology and its Department of Virology were parts of the Sarajevo Faculty of Medicine until 1974. A major process of integration of healthcare institutions in the Sarajevo area took place during the 1974-1978 period. BOAL Institute of Microbiology "Dr. Robert Fried" was thus established within the COAL UMC Sarajevo. Microbiological laboratories located at "Jezero" and "Koševo" and the Institute of Dermal and Venereal Diseases "Dr. Simo Milosevic" at Nahorevo were also merged. The Institute of Microbiology of the COAL UMC was given the legal status of an independent organization. Over the time the Institute of Microbiology "Dr. Robert Fried" was renamed into the Institute of Microbiology, Parasitology and Immunology "Dr. Robert Fried" and it existed as such until organizational changes in 1992, when it was transferred to the Koševo hospital premises. In 1974 Prof. Zivorad Miladinovic was appointed Director of the Institute, who was subsequently succeeded by Dr. Marija Miladinovic and Prof. Mira Nadazdin, respectively. The integration was carried out with the aim to gather the professional and scientific staff into a single institution, and the arrival of several prominent experts contributed to even more successful work and development of the Institute. Several prominent experts in certain fields of microbiology, such as Prof. Mira Nadazdin, Dr. Ksenija Karlovac, Dr. Antonija Semze-Rafaeli, Dr. Ljubica Radonjic, Dr. Medzida Hadzijamakovic, Dr. Kornelija Dimitrijevic, Dr. Nada Lucic, Prof. Fahrija Basic, Jovanka Vidakovic, BS in Biology, Dr. Maja Bratic, Prof. Kemal Seric, Dr. Gordana Jovanovic, Dr. Biljana Pesic-Pecek, Dr. Sukrija Zvizdic, Dr. Zoran Pikula, Dr. Gordana Pikula, Dr. Bojana Kosoric, Dr. Ljiljana Kordic, Dr. Mensura Seremet, Dr. Zvonka Knezevic, Dr. Jasna Jusupovic, Dr. Dijana Beus and others, made a significant contribution to the work of the Institute. In the 1974-1987 period curricular, scientific and professional activities were being performed, and all equipment, premises and staff of the Institute were at disposal for the purpose of education. At this time, the Institute engaged Dr. Vera Busic who after defending her dissertation was promoted to Assistant Professor, and remained working at the Institute until 1981.

For the purpose of education, in 1974 the Institute employed assistants Fahrija Basic, BS in Biology, Dr. Svjetlana Sabovljev and Dr. Slavica Milutinovic. Dr. Edina Beslagic and Dr. Gordana Jovanovic were appointed as assistants in 1976, and Danica Hlaca and Zivorad Miladinovic were promoted to assistant professors, while Dr. Vera Busic successfully defended her dissertation.

Thanks to the staff improvements the work on publication of educational textbooks and teaching aids continued at the Institute. Accordingly, in 1976 assistants Dr. Vera Busic, Jovanka Vidakovic, Dr. Svjetlana Sabovljev and Dr. Slavica Milutinovic published the first Workbook in Microbiology for students of the Sarajevo Faculty of Medicine.

During this period of the Institute of Microbiology development, Prof. Dr. Danica Hlaca was the Head of the Virology Department. After successfully defending her dissertation at the Sarajevo Faculty of Science in 1979, Assistant Professor Danica Hlaca was promoted to Associate Professor in 1982. In 1987 she was promoted to Full Professor and subsequently in 1996 she retired as the Head of the Department of Microbiology of the Faculty of Medicine University of Sarajevo.

In order to keep the scientific and educational processes at the Institute on the highest possible level significant efforts were made in the upcoming period, which resulted in the engagement of veterinarian Prof. Kemal Seric, at the Virology Department of the Institute of Microbiology in 1977. His task was to establish a laboratory for production and maintenance of cell cultures, and a laboratory for isolation and identification of viruses. Over the time he successfully accomplished his task by enabling the isolation and typing of certain viruses, pathogenic to humans and animals, to be performed in Sarajevo until 1992. He completed his specialization at virology laboratories in Ljubljana, Zagreb and Belgrade and established close cooperation with prominent experts from these cities, maintaining continuous contacts and establishing a permanent system of virology laboratory visits. He paid special attention to education of younger staff and publication of scientific research papers in the country and abroad. In 1980s, Prof. Kemal Seric published two scripts in the field of microbiology for students of the School of Health Sciences, while Assistant Professor, Mira Nadazdin participated as an associate in the publication of Yugoslavian edition of the textbook "Microbiology and Parasitology" with Prof. Dr. Bogdan Karakasevic as an editor-in-chief.

In 1978 Prof. Mira Nadazdin, BS in Biology, was appointed Director of the Institute of Microbiology and she remained at this function until 1992, the beginning of aggression on the state of Bosnia and Herzegovina and wartime events in Sarajevo, when she left the Institute and the city.

In the 1978-1992, in the process of education within the course of Microbiology and Parasitology the Institute engaged the following assistants: Dr. Aleksandar Bogdanovic, Dr. Biljana Pecek, Dr. Sukrija Zvizdic, Dr. Bojana Kosoric and Dr. Ljiljana Kordic. Assistant Professor, Zivorad Miladinovic, took over a new position at the Faculty of Medicine in Banja Luka, while Mira Nadazdin, PhD was promoted to Assistant Professor at the Faculty of Medicine University of Sarajevo in 1981.

In 1985, Prof. Fahrija Basic was engaged at the Faculty of Pharmacy University of Sarajevo as Assistant Professor who in 1991 was promoted to Associate Professor and subsequently retired in 1997. For the purpose of education at the College of Health Sciences University of Sarajevo (which was transformed into the School of Health Sciences and Faculty of Health Sciences), Prof. Kemal Seric was appointed as Professor at the College and subsequently at the School of Health Sciences. In 1987 Dr. Sukrija Zvizdic was appointed as Assistant and was promoted to a lecturer and Assistant Professor at the School in 1990 and 1993, respectively.

After completing his specialization in Microbiology and Parasitology (1984), Dr. Sukrija Zvizdic established several virology laboratories within the Virology Department and also took over the position of the Head of the Virology Department. He established the laboratory for diagnosis of viral causal agents of gastrointestinal infections in humans, laboratory for production of viral antigens and immune serums, and also the laboratory for isolation of viruses from water. He continued his professional development in the field of virology at the School of Public Health "Dr. Andrija Stampar" and Medical School of the Zagreb University, and also at virology laboratories in Ljubljana, Belgrade and Lausanne, Switzerland. He also participated in number of symposiums, meetings and congresses of virologists and microbiologists from Yugoslavia and the world. At the same time, Dr. Ljiljana Kordic established the laboratory for diagnosing chlamydia and enterotoxin of some of bacterial species using the cell culture method, and for serological diagnosing of viral diseases. She visited virology laboratories in the USA and Europe. Ljilja Ivanjc, BS in Biology, was engaged with a task to introduce certain number of different types of cell cultures, necessary for isolation and typing of viruses, into the regular professional and scientific work of the Institute. In this period a close cooperation with prominent virology experts from the former Yugoslavia was established. Special attention was given to the education of employees, their participation in professional and scientific activities, as well as to the publication of professional and scientific papers in the country and abroad.

At the end of the 20th century at the Department of Microbiology special attention was given to the publication of student textbooks and workbooks. In that regard, Prof. Dr. Fahrija Basic and Prof. Dr. Edina Beslagic published the university textbook "Microbiology - Morphological Aspects and Diagnostics", followed by the university textbooks "General Medical Virology", "Medical Virology" and "Virology" published by Prof. Dr. Sukrija Zvizdic. Since 1995 Prof. Dr. Edina Beslagic, Prof. Dr. Fahrija Basic and Prof. Dr. Sukrija Zvizdic have prepared number of editions of the Workbook for students of the Faculty of Medicine, Faculty of Dentistry, Faculty of Pharmacy and School of Health Sciences. Applying the same principle of the Workbook publication, Prof. Dr. Sukrija Zvizdic, Prof. Dr. Edina Beslagic and Prof. Dr. Elvedina Kapic (2006) published the Manual for Students of the Faculty of Pharmacy, while Prof. Dr. Sukrija Zvizdic and Prof. Dr. Edina Beslagic (2007) published the Workbook in Microbiology for students of the School of Health Sciences - course of Medical Laboratory Diagnostics, for practical classes in the process of education. Prof. Dr. Edina Beslagic et al. published a faculty edition of "Medical Microbiology" in 2010, while Prof. Dr. Sukrija Zvizdic and Prof. Dr. Sadeta Hamzic published the university textbook "Medical Microbiology" for students of the Faculty of Pharmacy in 2015.

Numbers of specialized exams were organized at the Institute since its establishment until 1992. The Institute itself became a backbone of medical microbiology development in Bosnia and Herzegovina and the former Yugoslavia. In this period, a number of employed physicians and other experts with university degrees attended the specialized course and successfully passed the specialist exam in the field of medical microbiology and parasitology. Also, a number of experts with different degrees continued their professional development in well-known centers across the world, thus acquiring knowledge necessary for establishing microbiological laboratories of different profiles. Until 1992, majority of specialized students or specialists attended and successfully completed

their postgraduate studies at the Faculty of Medicine University of Sarajevo, School of Public Health "Dr. Andrija Stampar" and Faculty of Medicine University of Zagreb. Assistants Dr. Edina Beslagic and Dr. Gordana Jovanovic passed the specialist exam in Microbiology and Parasitology, and subsequently continued their education and successfully completed their postgraduate studies at the Faculty of Medicine University of Sarajevo. Dr. Bojana Kosoric and Dr. Ljiljana Kordic took the same path. At the same time, Dr. Biljana Pesic and Dr. Sukrija Zvizdic attended the specialized course in Microbiology and Parasitology at the Institute in Sarajevo and Faculty of Medicine University of Zagreb, and subsequently passed their specialist exams in Microbiology and Parasitology at the Zagreb Faculty of Medicine in 1984. At the same time, they attended and successfully completed their postgraduate studies in Medical Microbiology, the only one of this kind in Yugoslavia, at the School of Public Health ¨Dr. Andrija Stampar¨ and the Zagreb Faculty of Medicine. Following the postgraduate studies, Dr. Edina Beslagic, Specialist in Microbiology and Parasitology, Dr. Gordana Jovanovic-Zurovac, Specialist in Microbiology and Parasitology, and Dr. Ljiljana Kordic, Specialist in Microbiology and Parasitology, successfully wrote and defended their master theses before the commission of the Faculty of Medicine University of Sarajevo, while Dr. Sukrija Zvizdic, Specialist in Microbiology and Parasitology, successfully wrote and defended his master thesis before the commission of the Faculty of Medicine University of Zagreb, and obtained the degree of Master of Medical Sciences. Afterwards, Dr. Edina Beslagic, M. Sc. and Dr. Sukrija Zvizdic, M. Sc. successfully defended their dissertations at the Faculty of Medicine University of Sarajevo, obtaining the degree of Doctor of Medical Sciences. They were subsequently promoted to Assistant Professors, Associate Professors and Full Professors at the Department of Medical Microbiology, Parasitology and Virology of the Faculty of Medicine University of Sarajevo.

From 1987 until the closure of the Virology Department of the Institute of Microbiology, Parasitology and Immunology "Dr. Robert Fried" in Sarajevo, Prof. Sukrija Zvizdic, worked as the Head of the Virology Department, Specialized in Medical Microbiology and Parasitology, Master of Medical Sciences and Doctor of Medical Sciences. Due to organizational changes at the end of 1980s, the Virology Department was transferred to the premises of the Institute of Microbiology, Parasitology and Immunology "Dr. Robert Fried" at Nahorevo, where it remained until 1992 and the beginning of wartime events (closure of the Department). Prof. Sukrija Zvizdic, Prof. Kemal Seric, Jovanka Vidakovic, BS in Biology and Specialist in Microbiology and Parasitology, Dr. Ljiljana Kordic, Dr. Branka Vidakovic, Ljilja Ivanjc, BS in Biology, five laboratory assistants, two laboratory glassware washers, one cleaning lady and one worker in the room for experimental animals worked at the Department from the moment it was transferred until 1992.

The Department of Microbiology and Parasitology existed within the Institute of Microbiology, Parasitology and Immunology "Dr. Robert Fried" in Sarajevo until 1987, when it was separated from the Institute and merged with the Faculty of Medicine University of Sarajevo. As such, the Department was still connected to the Institute in organizational and professional terms. The process of education was organized at the Department of Microbiology and Parasitology for students of the Faculty of Medicine, Faculty of Dentistry, Faculty of Pharmacy, and mostly for students attending particular courses at the School of Health Sciences.

The Institute of Microbiology, Parasitology and Immunology "Dr. Robert Fried" was transferred from the Faculty of Medicine to newly-built special premises at Kosevo in 1992, when it was merged with the Clinical Center University of Sarajevo. As a consequence of the relocation and the 1992 war events in Bosnia and Herzegovina the Institute was reorganized and rearranged. Prof. Dr. Sukrija Zvizdic, served as the Acting Director of the Institute for a short period of time, after which Prof. Kemal Seric, was appointed the Director of the Institute within the CCUS.

Since its establishment until its transfer and merger with the CCU Sarajevo in 1992, major professional and organizational changes were evident, as well as the efforts of the Institute to adapt to a newly incurred situation and to keep expanding its activities and increasing the number of employees with more or less success. All of this was aimed at providing the patients with the improved diagnostic services and the employees with quality professional development, in order that they could transfer their knowledge to students and younger colleagues. In 1975 the Institute of Microbiology had 53 employees, increasing to 107 in 1986 and reaching 112 in 1992. Out of the total of 112 employees, majority of them were laboratory technicians, whereas 30 employees had university degrees. Those were physicians attending the specialized course and physicians - Specialists in Medical Microbiology and Parasitology, Bachelors of Biology - Specialists in Medical Microbiology, Bachelors of Biology, veterinarians and Bachelors of Law. In 1992, Prof. Ksenija Karlovac, Prof. Fahrija Basic, Prof. Edina Beslagic, Prof. Sukrija Zvizdic, and DVM, Kemal Seric obtained scientific degrees of Doctors of Medical Sciences, while Prof. Danica Hlaca, and Mira Nadazdin, obtained degrees of Doctors of Science in Biology. Two physicians - specialists, Dr. Gordana Jovanovic-Zurovac and Dr. Ljiljana Kordic, obtained degrees of Masters of Medical Sciences. All other physicians were attending the specialized course or they had passed their specialized exam in Medical Microbiology and Parasitology, while one physician had passed the specialized exam in Immunology.

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Ambroise Paré: «The father of modern surgery»

Ambroaz Pare: «Otac moderne hirurgije»

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ABSTRACT

The thorough study of historical researches that reflect the radical appearance, the gradual formation, but also the multidimensional contribution of medical achievements, as if of their talented inspirers deterministically lead to the unhindered expression of irrefutable truths. Multifarious personality, tireless scholar, versed in the diachronic humanistic ideals French Ambroaz Paré was able to confront seemingly insurmountable obstacles: his bad social status, the large financial difficulties, and additionally the unbearable everyday life on the battlefields of the endless conflicts which neither limited his fiery spirit nor put out his unparalleled desire for substantial solidarity and selfless love for all human beings. On the contrary, these parameters forged a magnanimous physiognomy that is reasonably admired for its scientific foundation, the unique inventiveness and the rich authorial activity. Furthermore, the modest doctor barber impressed the mankind by his rare moral mosaic and the unpretentious faith in God. Based on the mentioned facts Pare obtained the universal acceptance of the medical community that awarded him with the superb title of "father of modern surgery". However all generations constantly pore over his enlightened thoughts and receive valuable material related to surgery, obstetrics, forensic science and pharmacology.

Key words: Paré, surgery, obstetrics, forensic, pharmacology

SAŽETAK

Temeljno proučavanje istorijskih istraživanja koje odražava radikalan nastanak, postepeno formiranje, ali i višedimenzionalan doprinos medicinskih postignuća, kao i njihovih obdarenih inspiratora deterministički vodi do neometanog izražavanja neoborivih istina. Raznovrsna ličnost, neumorni naučnik, upućen u dijahronične humanističke ideale francuski Ambroaz Pare uspeo je da se suprotstavi naizgled nesavladivim preprekama: loš socijalan status, konstantne finansijske teškoće, ali i nepodnošljiva svakodnevnost na krvoločnim ratištima beskrajnih konflikta niti su ograničili njegov vatren duh niti mu ugasili neuporedivu želju za suštinskom solidarnosti i nesebičnom ljubavi prema svim ljudskim bićima. Nasuprot tome, ti parametri iskovali su velikodušnu fizionomiju kojoj se razložno divi radi naučne podloge, jedinstvene inventivnosti, ali i bogate autorske delatnosti. K tome skroman lekar berberin impresionira čovječanstvo svojim retkim moralnim mozaikom i nepretencioznom vjerom u Boga. Na osnovu navedenih činjenica Pare je stekao univerzalno prihvatanje medicinske zajednice koja ga je nagradila vrhunskim imenovanjem „oca savremene hirurgije". Međutim sve generacije stalno se osvrću na njegove prosvetljene misli i primaju dragocjeno gradivo u vezi sa hirurgijom, akušerstvom, forenzikom i farmakologijom.

Ključne reči: Pare, hirurgija, akušerstvo, forenzika, farmakologija

BIOGRAPHICAL DATA

Ambroise Paré was a French military doctor, anatomist and inventor of surgical instruments. Due to his multifaceted intellectual personality and his polyhedral presence in general he is considered one of the pioneers of surgery and forensic pathology.

He was born around 1510 in the village of Bourg-Hersent, near Laval, the capital of the Mayenne department. His unfailing love for learning and his impressive ingenuity were unfortunately not rewarded with valuable education because of miserable economic conditions of his family (his father was a barber). Also, his authentic inclination for professional occupation with medicine was also hampered by the dominated perception of the time that surgery was accounted for an undignified, «impure» job and therefore only the so-called «barber surgeons» could be engaged in it.

Given that his father and also his brother were barber surgeons, Paré was their apprentice who quickly became a capable professional. At the age of thirty he began his activity at Laval where his father worked as the earl's personal barber surgeon.

However, his sparkling wit and his personal ambitions remained unfulfilled by the everyday monotonous work in the small provincial town. Therefore, he decided to travel to Paris in order to improve his knowledge in surgical practice: from 1532 to 1536 he worked as an assistant and nurse in the hospital «Hotel Dieu», while he was taught anatomy and surgery. Paré characterized that period as the most important for his whole future career.

Unfortunately his ardent desire of getting the official license for being a barber surgeon seemed to be unrealizable, since he couldn't pay the required fees. So he left the French capital and decided to be trained in surgery through his active participation in actual military conflicts.

In the battle of Piedmont (1536), during King Francis II's campaign, Paré was confronted with a shocking experience which determined his whole attitude: two soldiers were injured by gunpowder; their partners asked him if he could relieve the wounded comrades from terrible pains. When Paré replied that it was impossible, without hesitation the determined soldier rushed and cut their throats wishing such a redemptive treatment to be used in any identical circumstances.

From 1537 to 1539 he was attending military operations of lieutenant general René de Montjean. After Montjean's death (1539) he returned to Paris. Having the necessary financial sources he managed to pass the exams and obtained the license of a barber surgeon. In 1552 he became a member of the royal medical staff of the Valois Dynasty under Henry II. In 1563 he was appointed director of the Surgical Department in the «Hotel Dieu» hospital. Subsequently he was appointed the king's surgeon and held that titular position until his death.

During his multiannual career he served as personal surgeon of four kings (Henry II, Francis II, Charles IX και Henry III). In fact, he attended Henry II's tragic end (10 July 1559): during a joust he was killed by Gabriel Montgomery, captain of the King's Scottish Guard.

On a personal level, in 1539 he married Jean Mazeline, a vintner's daughter who gave him three children. In 1573 his wife died; in 1574 he married Jacqueline Roussel with whom he had six children. Provided that he was a Huguenot as a member of the Protestant Reformed Church, on Saint Bartholomew's Day Massacre (24 August 1572) Paré was liable to being strung up by the raging crowd. King Charles IX locked him in a closet, so the famous surgeon was rescued. At the age of eighty, on 20 December 1590, he died in Paris from natural causes and was buried with great honor in the church Saint-André-des-Arts.

PARÉ'S INCOMPARABLE CONTRIBUTION TO MEDICINE

In the view of selfless given to every distressed fellow man, and especially to the numerous wounded in the bloodthirsty battles Paré was distinguished for his pluralistic involvement in any manifestation of medical science:

Surgery – Traumatology

The widespread use of firearms and muskets which replaced the mainly treatable sword wounds resulted in massive emergence of large open wounds and broken bones which required new methods of scientific (practical and theoretical) approach.

Many physicians of that time were applying precarious surgical methods and unsafe ways that very often caused painful death or exhaustion of already overburdened patients: castration in hernia operations; closing of the large arteries with hot iron during limb amputation; treatment of wounds using boiling oil and fat; intentional inducement of bleeding or vomiting; violent sweats. With despair Paré was observing the daily application of such disappointing and fatal alternative treatments and considered the development of innovative techniques as an urgent need and his moral obligation.

In his autobiographical work «Journeys in Diverse Places» we witness his first impressive success, as he made a comparison on the healing progress between two groups of patients: 1) those who had been submitted to the traditional therapy based on boiling oil of berries and cauterization; 2) those to whom he had applied an enterprising receipt of egg yolk, rose oil and mineral turpentine. The former still suffered from terrible pains, the latter who had experienced the old Roman technique with depressant patches avoided the cauterization of wounds thanks to the antiseptic properties of turpentine.

It is worth mentioning that Paré manifested a strong tendency to be differentiated from the traditional medical structures. More specifically, he was the first who replaced the practice of cauterization and connected syphilis with the aortic aneurysm. Using the method of blood vessels ligation at the limb amputation he was able to reduce the bleeding incidents, but not to prevent the inevitable expansion of infections. In addition, he deployed the compressions, introduced the implantation of the dental gold or silver and the use of prosthetics (artificial limbs and eyes) made by enameled gold, silver, porcelain and glass.

Not only was Paré meritorious for offering surgical relief from the severe pain after amputation operations, but he was also closely interested in the scientific clarification of that phenomenon: he expressly emphasized that this kind of pain was based in brain. Furthermore he designed the groundbreaking haemostatic technique «Bec de Corbeau» (Crow's beak) which he thoroughly described in his book «Treatise on Surgery» (1564). Reasonably, this method became the favorite method of the wounded soldiers whose treatment was so unpleasant until then, due to the fact that it implied a serious disability or a horrible outcome.

Obstetrics

Apart from being a leading personality of surgical science in the 16th century Paré renewed obstetrics in two ways. Firstly, he improved podalic version: turning the fetus within the womb and bring it to a position suitable for a smooth birth. Secondly, in cases of excessive bleedings due to premature birth he averted the cruel fetal partition and its violent extraction piece by piece.

Even in infants who had difficulties during teething he used to apply the piercing of their gums with a scalpel, since he believed that the teeth that did not rise could often result in death.

Forensic Medicine – Pharmacology

With systematic writing of «Reports in Court» reflecting legal considerations and basic instructions with medical background, Paré founded the modern Forensic.

He was also involved in direct dealing with poisoning. Already in 1567 he proved the non-absolute effectiveness of bezoar stones that were believed to be panacea. One of his cooks sentenced to be hanged for theft was suggested to be released in case he survived swallowing poison and getting a bezoar stone at once. The cook died after many hours of indescribable physical torture.

Works

In 1575 Ambroise Paré published all his scientific works from which the most noticeable are:

- Methods for treating wounds caused by arquebus and firearms (1545);
- Journeys in Diverse Places (1552);
- Therapeutic method of wounds and fractures of human head with images of organs (1561);
- Treatise on Surgery (1564);
- On plague, smallpox and measles with a brief description of leprosy (1568);
- On creation of man and monsters (1573).

CONCLUSION

Given that Paré tried to systematically upgrade the invasive medicine to a recognized professional level presented a real reformer of the Renaissance surgery. Along with the continuous scientific observations, the authentic piety and the exemplary humanism he managed to redeem wounded soldiers from unbearable pains or an undesirable death. Therefore we realize that Ambroise Paré deserves his prominent position among the most noteworthy benefactors of humanity.

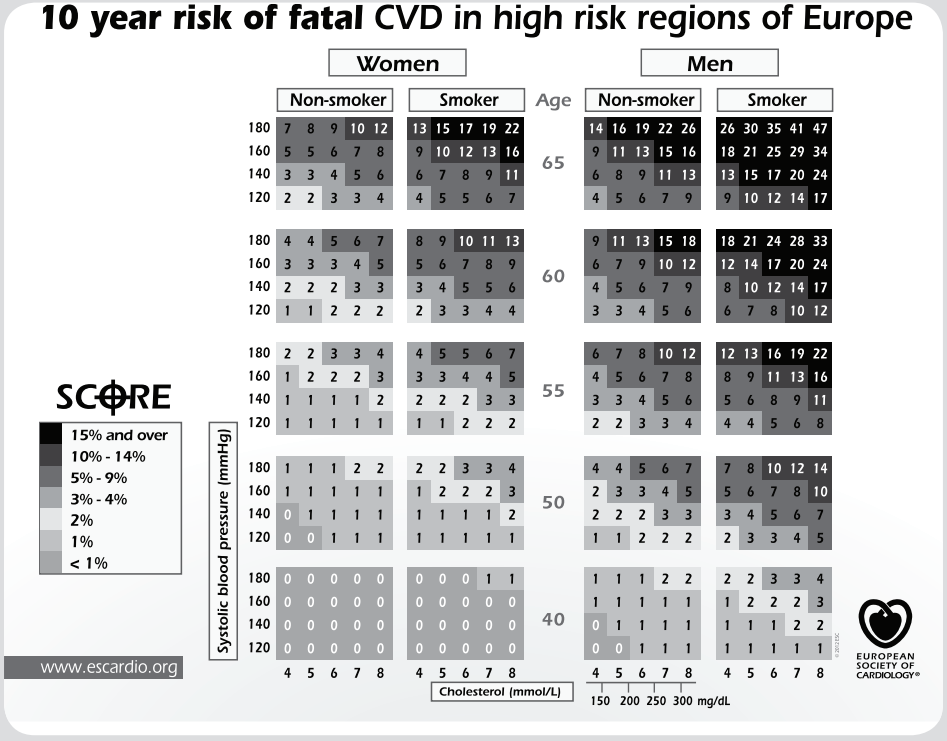
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Ischemic stroke caused by carotid dissection

Ishemijski moždani udar uzrokovan disekcijom karotidne arterije

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ABSTRACT

Craniocervical artery dissection is a rare nonatherosclerotic vascular disease of unknown cause and the cause of ischemic stroke in young adults and middle aged people. The prevalence of extracranial carotid dissection in the general population is underestimated due to asymptomatic forms of the disease and the nonrecognition. It is considered that the craniocervical artery dissection caused focal brain ischemia in 1-4% of all patients. Its clinical picture varies and the manifestations range from mild cerebral dysfunction and/or lesions cranial nervs to brain infarction. In suspected extracranial dissection and after the brain CT thorough ultrasonographic and angiographic analysis follows. Our article presents a case of a patient with signs of fresh brain ischemia caused by spontaneous dissection of the internal carotid artery.

Key words: spontaneous dissection of craniocervical arteries, angiographic analysis, ischemic stroke

SAŽETAK

Disekcija kraniocervikalnih arterija rijetka je neaterosklerotična vaskularna bolest nepoznate etiologije i uzrok je ishemijskog moždanog udara kod mladih osoba i onih srednjeg uzrasta. Učestalost ekstrakranijalnih karotidnih disekcija u opštoj populaciji je potcijenjena zbog asimptomatskih oblika ali i neprepoznavanja bolesti. Danas se smatra da je disekcija kraniocervikalnih arterija uzročnik žarišne ishemije mozga u 1-4% svih oboljelih. Njena klinička slika je varijabilna, a manifestacije se kreću u dijapazonu od blage moždane disfunkcije i/ili lezije moždanih živaca do slike infarkta mozga. Kod sumnje na ekstrakranijalnu disekciju poslije CT pregleda mozga slijedi obimna ultrazvučna i angiografska analiza . U radu je prikazan slučaj pacijentice sa znacima svježe ishemije mozga uzrokovane spontanom disekcijom unutrašnje karotidne arterije.

Ključne riječi: spontana disekcija kraniocervikalnih arterija, angiografska analiza, ishemijski moždani udar

INTRODUCTION

Spontaneous dissection of craniocervical (carotid and vertebral) artery (DCA) is rare nonatherosclerotic vascular disease of unknown cause and the cause of ischemic stroke in young adults and middle aged people. A rare cause of stroke in the general population (1-4% of all patients), but more common in patients younger than 30 years (5%-20%), slightly more in women (1). The first published work of spontaneous dissection of the internal carotid artery was published in 1954, and since then interest in this disease as a cause of focal brain ischemia, in continuous growth. In most cases, the cause and pathogenesis of aortic disection not clear, but it is generally divided into two groups: traumatic (after head and neck injuries resulting in injury to the arteries) and spontaneous. According to the localization of the intra- and extracranial. It is conducted a multicenter, randomized study on the directions of cervical arteries and patients with ischemic stroke (Cervical Artery Dissections and Ischemic Stroke Patients - CADISP), which analyzed risk factors and association with the emergence of DCA stating among other things, recent surgery on the head or neck as the cause of DCA (2). The pathogenesis of this disease remains unclear. Pathological, craniocervical artery dissection is manifested in the form of hematoma in the wall of arteries (carotid or vertebral), which occurs as a result of secondary cleft intimacy or directly caused by bleeding within the arterial wall caused by the rupture of vasa vasorum. Intramural hematoma can spread to the intima or adventitia, resulting in stenosis or aneurysmal dilation of the arteries. The mechanism by which DCA causes ischemia of the brain or the retina in most cases is the embolism intraluminal thrombus that forms at the site of cleavage of intimacy (2,3). Hemodynamic infarction is also possible. The development of new diagnostic methods DCA gain increasing importance in the genesis of brain ischemia, especially in younger patients. Clinical manifestations of the arteries of the neck dissection can be very varied and constitute a major problem with differentially number of neurological diseases such as occlusive vascular disease, tumors of the nervous system, migraine headache and others. but also with other diseases type cervical syndrome, ear infections (4).

Our article presents a patient with signs of fresh brain ischemia caused by spontaneous dissection of the internal carotid artery.

CASE REPORT

A 52 years old patient was hospitalized at the Neurology Clinic for further diagnostic procedures due to paralysis of the left body side. Present illness occurred 20 days before the admission, when the patient suffered from headache and high blood pressure. The patient was examed by a competent internist who included antihypertensive therapy (newly discovered hypertension). Three days later the patient reported weakness of the left leg and was hospitalized in other medical institutions where a CT scan of the head showed hypodense which

represented post ischemic focus, although other etiology was not excluded. The patient was referred to our clinic for additional clinical trial. The personal history of a newly discovered hypertension required regular therapy. In the last three months of hypothyroidism Letrox was included in the therapy and the left lobe of the thyroid gland was surgically removed. There were no other risk factors for cerebrovascular disease. The patient is married with two children, employed in an elementary school, non smoker or drinker. On admmission the patient was cardiorespiratory compensated, normotensive, orderly complete somatic findings. Neurological examination recorded left cerebral hemiplegia with minimal movement in her left hand with a positive Babinski left. NIHSS 7. The offices of the complete laboratory (CBC, SE, minerals, transaminase, urea, creatinine, Blood sugar, acid uricum, total bilirubin, Fe, total protein) in referal valves. Status of lipids: cholesterol 5.0 triglycerides 1.22. Thyroid hormones (TSH, FT3, FT4) in ref. values. EEG and X-ray of the chest organs were normal. Cardiology review showed: ECG rhythm sinus, fr 77 / min, with no signs of ischemia, lesions and myocardial necrosis. DG: Cor compensatum, Hypertension arterialis and ECHO normal findings.

MRI kranium (30.05.14) (Figure 1) indicated a fresh lesion right in the ventral part of the nucleus lentiformis, front arm and the root of the internal capsule, caudate nucleus caput and frontoparietal supraventricular mostly parasagital subcortical and cortical with hemorrhagic transformation.

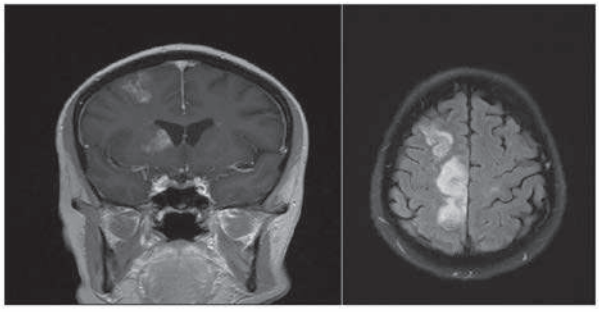


Figure 1 MRI craniuma indicates a fresh lesion.

In 3D TOF MRA (Figure 2) thrombus on a long section in the final part of the extracranial internal carotid artery, and petrosal initial segment of the internal carotid artery on the right side can be seen on the ground before dissection, which is now thrombosed.

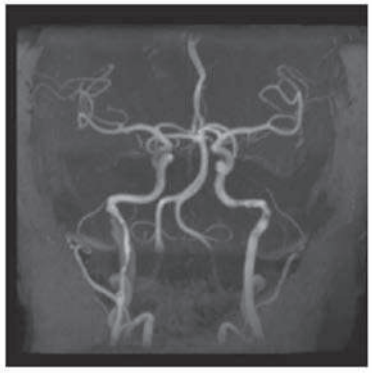


Figure 2 Thrombus on a long section in the final part of the extracranial ACI on the ground before dissection.

CTA blood vessels of the head and neck and aortic arch (06.06.2014) after i.v. administration of 100 ml contrast. Right ACI gracile of level crossings C3, C2 with visible dissection at the level of C1-C2 distally around the contours of thrombus semilunaris to enter the petrous part of the temporal bone represents probably thrombosed false lumen. Vascular surgeon examining the MRA and CTA (thrombus in the lumen and pseudolumen ACI) did not set indications for surgical treatment. The patient was treated with clopidogrel, antihypertensive, lipid lowering drugs, analgesics, and infusion solutions. The antibiotic were administered for urinary infection and physical therapy was carried out at all times. After the therapy the patient was somatically stable, conscious, afebrile, cardiorespiratory compensated, normotensive, with urinary infection in sanation. Following discharged, the neurological findings showed the left cerebral hemiplegia crural type. NIHSS = 6.

Control CTA blood vessels of the head and neck and aortic arch (20.07.2016 y.) (Figure 3) after application of 100 ml contrast i.v.: right ACI diameter up to 4.7 mm Neat. Characters dissection not visible on check-up, visible goods protrusion lateral contour at the level of i.v. the area C1 / C2 can be in terms of lower PAH lesions. Control neurological examination (12.08. 2015) indicates discrete left hemiparesis cerebral, self-mobile patients. NIHSS 2.



Figure 3 Characters of dissection not visible on check-up.

DISCUSSION

Dissection of the extracranial carotid artery is the cause of ischemia, more often in younger people manifesting acute neurologic deficit that may be permanent or transitory (4). In most cases, the cause and pathogenesis of aortic is not clear, but it is generally divided into two groups: traumatic (after head and neck injuries resulting in injury to the arteries) and spontaneous and based on localization of the intra - and extracranial (5). According to a multicenter, randomized studies on the directions of cervical arteries and patients with ischemic stroke (Cervical Artery Dissections and Ischemic Stroke Patients - CADISP), which analyzed risk factors and association with the occurrence of DCA among other listed recent surgery on the head or neck as the cause of DCA (2). Three months prior to these neurological symptoms our patient underwent the neck surgery, when the left lobe of the thyroid gland was removed, which could be considered the cause of carotid artery dissection likely due to trauma. Expansive hematoma

(intramural hematoma - IMH) in the blood vessel wall is the basic lesion during dissection, which can evolve in different ways, leading to different pathoanatomic substrate. Subintimal localized IMH due to the increase caused by stenosis with apparent thrombus formation or primary occlusion, as was the case with our patient (2). Predilective place for the development of carotid artery dissection is distal cervical segment. Mutual (spontaneous) carotid dissection are rare and occur in 5-10% of cases (6). The diagnosis was confirmed based on ultrasound screening and magnetic resonance imaging with cervical magnetic resonance (MRA) and computed angiography (CTA) (7). In our patient we confirmed dissection of the internal carotid artery and the presence of pseudo thrombus with MRA and CTA, as well as complete lysis of thrombus control CTA. The anticoagulant or antiplatelet therapy is the recommended treatment in the acute phase of DCA in order to prevent the primary or re-ischemic event. The current best evidence is the Cochrane systematic meta-analysis of non-randomized studies that show the outcome of antiplatelet compared to anticoagulant therapy in the absence of randomized data (8). The proportion of patients with complete resolution of arterial abnormalities varies in studies and it is about 46% of stenosis, 33% of occlusion and 12% of dissecting aortic aneurysm in the general population (9). Complete recanalization was highest in patients with DCA who had only local Symptoms and signs (9). In the general population in the case of DCA complete resolution or stable residual lumen is documented after an average follow-up ranging from 0 to 29 years and in 82% of cases during the first year (9). In our case, the patient was treated with antiplatelet therapy and lipoid (clopydogrel and rosuwastatin) for a period of one year, after which the CTA showed no dissection or pseudotromb.

CONCLUSION

The spectrum of clinical manifestations of carotid artery dissection is variable. Carotid artery dissection is not always associated with brain infarction. Given the fact that dissection is one of the main causes of ischemic stroke in young adults, early diagnosis and treatment of dissections are of great importance. Ultrasound duplex scan is a useful screening tool. However, the diagnosis should be confirmed by magnetic resonance imaging, magnetic resonance angiography, or computed tomography angiography if magnetic resonance is contraindicated. The pathophysiology of dissection is still unclear. According to some authors, patients with carotid artery dissection may have a constitutional and partly genetically determined susceptibility of the arterial wall; and external factors, such as mild trauma or infection, trigger the development of dissection.

Conflict of interest: none declared.

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Diagnostic evaluation and treatment of an infant with PHACE syndrome

Dijagnostička evaluacija i tretman dojenčeta sa PHACE sindromom

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ABSTRACT

Introduction: a small subset of children with infantile hemangiomas (IH) has associated congenital anomalies. The best-known such association is PHACE (Posterior fossa malformations, Hemangiomas of the head and neck, Arterial anomalies, Cardiac defects and/or abnormalities of the Eye). Case report: we describe a female term neonate with PHACE syndrome consisting of a left-sided facial segmental hemagioma >5 cm and a posterior fossa malformation (large posterior fossa cyst and left cerebellar hypoplasia). Apart from facial hemangioma, the physical examination, neurological status, standard laboratory tests, abdominal and heart ultrasonography were normal. Brain ultrasonography showed anechogenic cyst in posterior fossa. Magnetic resonance imaging of the brain confirmed the diagnosis of Dandy-Walker spectrum. Since PHACE syndrome has very high incidence of assocciated cererbrovascular anomalies in order to estimate the stroke risk before the start of propranolol therapy magnetic resonance angiography was done. The stroke risk was estimated low and the propranolol therapy was initiated. It was well tolerated by the baby and hemangioma showed a significant reduction after one month therapy. Conclusion: in patients with PHACE syndrome because of high incidence of cerebrovascular anomalies, before the start of propranolol therapy, additionally to standard pretreatment assessment, magnetic resonance angiography and stroke risk assessment is needed.

Key words: infantile hemangioma, PHACE syndrome, stroke risk, magnetic resonance angiography, propranolol therapy

SAŽETAK

Uvod: mali dio djece sa infantilnim hemangiomima (IH) ima pridružene urođene anomalije. Najbolje poznata takva asocijacija je PHACE sindrom (akronim od eng. Posterior fossa malformations, Hemangiomas of the head and neck, Arterial anomalies, Cardiac defects and/or abnormalities of the Eye). Prikaz slučaja: ovo je prikaz ženskog terminskog novorođenčeta sa PHACE sindromom koji se sastoji od ljevostranog facijalnog segmentalnog hemangioma > 5 cm i malformacije stražnje lobanjske jame (velika cista stražnje lubanjske jame i ljevostrana cerebelarna hipoplazija). Osim facijalnog hemangioma, fizikalni pregled, neurološki status, standardni laboratorijski nalazi, ultrasonografija abdomena i srca su bili normalni. Ultrasonografija mozga je pokazala anehogenu cistu u stražnjoj lobanjskoj jami. Magnetna rezonancija glave je potvrdila dijagnozu iz Dandy-Walker spektra. S obzirom da PHACE sindrom ima vrlo visoku incidenciju pridruženih cerebrovaskularnih anomalija, da bi se procijenio rizik od moždanog udara prije započinjanja terapije propranololom učinjena je magnetna angiografija mozga. Rizik za moždani udar je procijenjen kao nizak i započeta je terapija propranololom. Beba je terapiju dobro podnijela i došlo je do značajne redukcije hemangioma nakon mjesec dana terapije. Zaključak: kod pacijenata sa PHACE sindromom zbog visoke incidencije cerebrovaskularnih anomalija, prije započinjanja terapije propranololom, uz standardnu procjenu, treba uraditi magnetnu angiografiju mozga i procjenu rizika za moždani udar.

Ključne riječi: infantilni hemangiom, PHACE sindrom, rizik za moždani udar, magnetna angiografija, terapija propranololom

INTRODUCTION

With incidence of 4 to 5%, infantile hemangiomas (IH) are the most common tumors of childhood (1,2,3). A small subset of children with IH exhibit associated congenital anomalies. One such association is PHACE (Online Mendelian Inheritance in Man 606519) (3). The term PHACE is an acronym referring to Posterior fossa anomalies, Hemangioma, Arterial lesions, Cardiac abnormalities/Coarctation of the aorta, and Eye anomalies. It was first described in 1996 by dr Ilona Frieden et al. (1).

Medical therapy for IH includes both topical and systemic administration of medications.

Agents with reported activity in treating IH include steroids, interferon α, and, recently, propranolol. The use of propranolol for the treatment of IH started in 2008 and since then propranolol has become the first-line medical therapy (4,5). The mode of action of propranolol in the treatment of IH is unknown and proposed mechanisms include vasoconstriction, and inhibition of angiogenesis (via suppression of VEGF-A and down-regulation of MMPs and interleukin, regulation of the renin-angiotensin system and inhibition of nitric oxide production) (4,5). Prior to propranolol therapy, because of possible adverse effects, pretreatment assessment is recommended. This includes a complete history, physical examination and electrocardiography. Some clinicians even prefer to have a cardiology consultation (6,7).

Propranolol induced variability in systolic blood pressure may increase the risk of acute ischemic stroke on the ground of arterial anomalies which are commonly associated with PHACE syndrome. That is why, in PHACE syndrome even more detailed pretreatment assessment, including magnetic resonance angiography of the head and neck and cardiac imaging is needed. Infants with PHACE syndrome considered to be at highest risk for stroke are those with severe, long-segment narrowing or nonvisualization of major cerebral or cervical arteries in the setting of inadequate collateral circulation, especially when there are coexisting cardiac and aortic arch anomalies. If imaging results place a patient into a higher risk category for stroke, consultation and co-management with neurology is appropriate.

CASE REPORT

We described a patient with PHACE syndrome including a left-sided facial segmental hemagioma >5 cm associated with a large posterior fossa cyst and left cerebellar hypoplasia.

The neonate was admitted to our department at the age of 10 days. After delivery, a left-sided facial segmental hemagioma was seen in otherwise normal female term newborn. Prenatal ultrasonography was unspecific. Following discharge from maternity hospital the baby was referred to our Clinic for further diagnostics after spending few days at home totally asymptomatic.

This neonate was born at 37 weeks gestation by a spontaneous delivery to a 30-year-old white female and weighted 3050 gs. The infant had Apgar scores of 9 and 10 at 1 and 5 minutes, respectively.

At the admission to our department, apart from facial hemangioma, the physical examination and neurological status of the baby was normal (Figure 1).



Figure 1 Left-sided facial segmental hemangioma.

The standard laboratory tests (C-reactive protein, total blood count, differential blood count, urine) were also normal. We conducted the ultrasound imaging of brain, heart and abdomen.

Heart ultrasonography showed patent foramen ovale, otherwise it was normal. Abdominal ultrasonography was normal. Brain ultrasonography showed anechogenic cyst in posterior fossa with maximum width of 2.22 cm (distance from occipital bone to brain surface) and hypoplastic cerebellum (Dandy-Walker spectrum). Magnetic resonance imaging of the brain confirmed the diagnosis of Dandy–Walker spectrum consisting of a large posterior fossa cyst communicating to fourth ventricle and left cerebellar hypoplasia. There was no hydrocephalus, midline anomalies or neuronal migration disorder (Picture 2). Magnetic resonance angiography showed a few arterial knee-like turn-downs located in both internal cerebral arteries, both meddial cerebral arteries and triplication of A2 segment of left anterior cerebral artery, but with no narrowing. Demonstrable vessels at the base of the skull were normal (Figure 2).

Accordingly, stroke risk in this infant was classified as low and we started propranolol therapy.

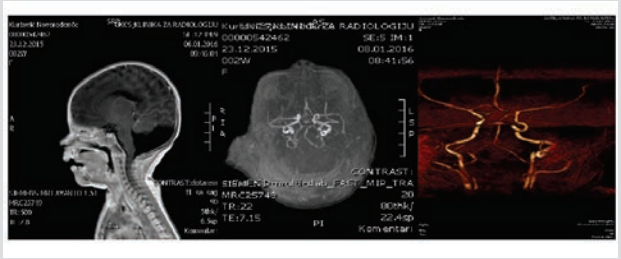


Figure 2 Sagital magnetic resonance image showing posterior fossa anomaly and magnetic resonance angiography images of cerebral blood vessels.

Following recommendations, we monitored heart rate and blood pressure at baseline and at 1 and 2 hours after receiving the initial dose, and after significant dose increase (>0.5 mg/kg/day). We started propranolol at the initial dose of 1 mg/kg/day. It was well tolerated and following protocol (), we gradually (0,5 mg/kg/day) increased it to a full dosage of 2 mg/kg/day. The baby was discharged from our department on 10th day of hospitalization. On discharge we recommended neuropediatric, cardiology and hematology follow up. On the first control, one month after initiation of propranol therapy, the baby was well and there was significant reduction of facial hemangioma (Figure 3).



Figure 3 Significant reduction of facial hemangioma after one month of propranolol therapy.

DISCUSSION

PHACE syndrome is now understood to be predominantly a congenital vasculopathy. The hallmark of PHACE syndrome is a large, segmental, often superficial IH, characteristically located on the face, scalp, and/or neck.

Cerebrovascular anomalies, present in more than 90% of patients, are the most common extracutaneous feature of PHACE syndrome, followed by cardiac anomalies (67%) and structural brain anomalies (52%) (1). The most common structural brain malformations are of cerebellum and posterior fossa structures such as Dandy–Walker complex, with or without cerebellar hypoplasia. The spectrum of anomalies in PHACE syndrome and the ipsilateral relationship between such anomalies and cutaneous IH strongly suggest a 'developmental field defect', whereby an insult at a critical time in embryogenesis gives rise to similar developmental outcomes (4). There are recent data suggesting that chromosomal region 7q33 may provide a genetic susceptibility to exhibit the PHACE phenotype.

The risk of PHACE syndrome in an infant presenting with a large, segmental IH (≥22 cm2) of the head or neck is approximately one-third (8).

A multidisciplinary group made a consensus on PHACE syndrome diagnosis in November 2008. Diagnostic critera were stratified into 2 categories: a) PHACE syndrome and b) possible PHACE syndrome (Table 1). PHACE syndrome includes facial hemangioma greater than 5 cm plus 1 major or 2 minor critera. The possible PHACE syndrome includes facial hemangioma greater than 5 cm plus 1 minor criteria or hemangioma of the neck or upper torso plus 1 major or 2 minor criteria or 2 major criteria without hemangioma (9).

Table I Diagnostic criteria for PHACE syndrome.

	Major criteria	Minor criteria
Cerebrovascular	Anomaly of major cerebral arteries Dysplasia of the large cerebral arteries Arterial stenosis or occlusion with or without moyamoya collaterals Absence or moderate-severe hypoplasia of the large cerebral arteries Aberrant origin or course of the large cerebral arteries Persistent trigeminal artery Saccular aneurysms of any cerebral arteries	Persistent embryonic artery other than trigeminal artery Proatlantal intersegmental artery (types 1 and 2) Primitive hypoglossal artery Primitive otic artery
Structural Brain	Posterior fossa anomaly Dandy-Walker complex or unilateral/bilateral cerebellar hypoplasia/dysplasia	Enhancing extra-axial lesion with features consistent with intracranial hemangioma Midline anomaly Neuronal migration disorder
Cardiovascular	Aortic arch anomaly Coarctation of aorta Dysplasia Aneurysm Aberrant origin of the subclavian artery with or without a vascular ring	Ventricular septal defect Right aortic arch (double aortic arch)
Ocular	Posterior segment abnormality Persistent fetal vasculature (Persistent hyperplastic primary vitreous) Retinal vascular anomalies Morning glory disc anomaly Optic nerve hypoplasia Peripapillary staphyloma Coloboma	Anterior segment abnormality Sclerocornea Cataract Coloboma Microphthalmia

In this case report we presented a neonate admitted to our department due to a left-sided facial segmental hemagioma >5 cm. Diagnostic workout showed a posterior fossa cyst, and a left cerebellar hypoplasia was also present. Therefore, we had the association of a large facial hemangioma with one major criterion, such as a posterior fossa anomaly. Following consensus criteria the diagnosis of PHACE syndrome was established. For large IHs with a risk of functional impairment or disfigurement systemic propranolol therapy is routinely initiated. In this case systemic propranolol therapy was also indicated, but with special considerations due to PHACE syndrome. These considerations were the consequence of extremely high incidence of cerebrovascular anomalies in PHACE syndrome. That is why prior to the initiation of propranolol therapy in patients with PHACE syndrome, additionally to standard pretreatment assessment, magnetic resonance angiography and stroke risk assessment should be done. If the stroke risk was small, propranolol therapy should be started.

If the potential benefits of propranolol outweigh the risks, a consensus group suggests that inpatient hospitalization be considered for infants 8 weeks of age or younger, preterm infants less than 48 weeks' postconceptional age, those with poor social support, and those with cardiac or pulmonary risk factors (10). The group recommended initiating therapy at a dose of 1 mg/kg per day, with escalation to a target dose of 1 to 3 mg/kg per day with dosing frequency of 3 times daily.

The most dramatic improvement using propranolol for IH occurs within 3 to 4 months of initiation of therapy. However, many investigators continue therapy until patients reach an age when IH would normally begin to regress without treatment, that is at least at 8 to 12 months of age (10).

CONCLUSION

The management of IHs has evolved considerably in the past decade. The therapeutic options for these tumors expended by discovery of the IH response to systemic β-blockers. Experience in the management of hundreds of infants with IH has shown propranolol to have an excellent safety profile and high tolerability (10). Pretreatment assessment including complete history and physical examination, with special attention to the cardiac and pulmonary systems is recommended. In patients with PHACE syndrome because of high incidence of cerebrovascular anomalies, additionally to standard pretreatment assessment, magnetic resonance angiography and stroke risk assessment is needed.

Conflict of interest: none declared.

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Introduction is a concise, short part of the article, and it contains purpose of the article relating to other published articles with the same topic. It is necessary to quote the main problem, aim of investigation, and/or main hypothesis which is investigated.

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- **REZULTATI**
- **DISKUSIJA**
- **ZAKLJUČAK**
- **LITERATURA**

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